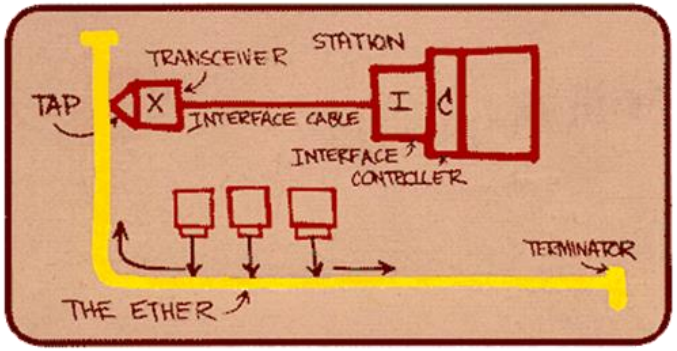
企業電腦網路





Chapter 5

Ethernet 802.3 LANs



中央資管

陳奕明

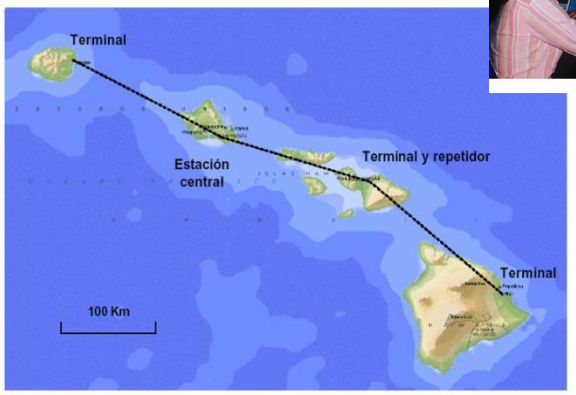
Ethernet Begins

- Introduction
- Physical Layer Standards
- Data Link Layer Standards
- Advanced Switch Operation
- Ethernet Security

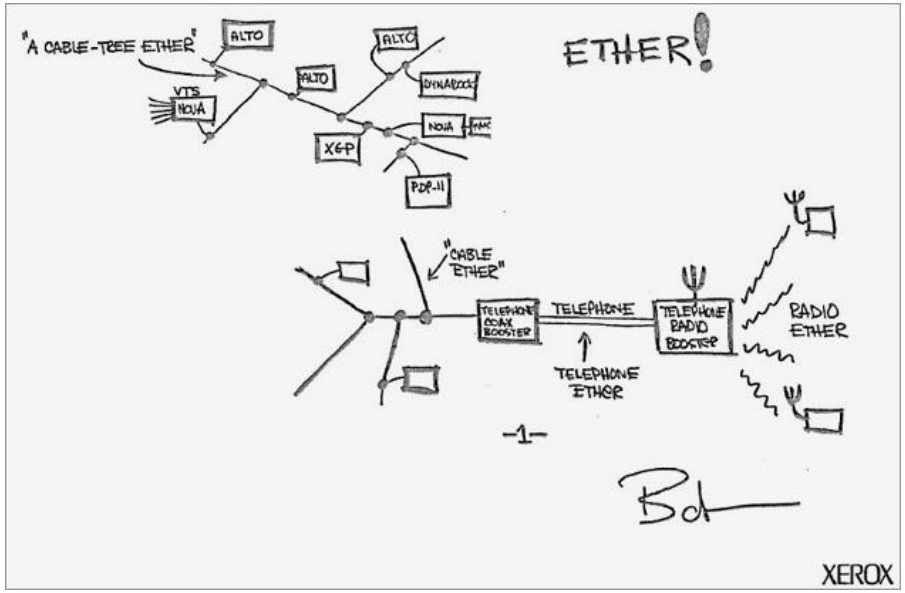
Bob Metcalfe

- Working on his PhD at MIT, then applied the **Xerox Palo Alto Research**
- visited the University of **Hawaii's ALOHANET** and developed a way to make it more efficient
- Applied it to a **cabled network**, called it **Ethernet** as a joke

Historia – Red AlohaNet







Ethernet Begins

Introduction

Physical Layer Standards

Data Link Layer Standards

Advanced Switch Operation

Ethernet Security

5.1 Simple Switched Ethernet Network(1/2)





Workgroup Switches Connect Hosts to the Network



Workgroup Ethernet Switch D



Workgroup Ethernet Switch E



Workgroup Ethernet Switch F



Client PC 1

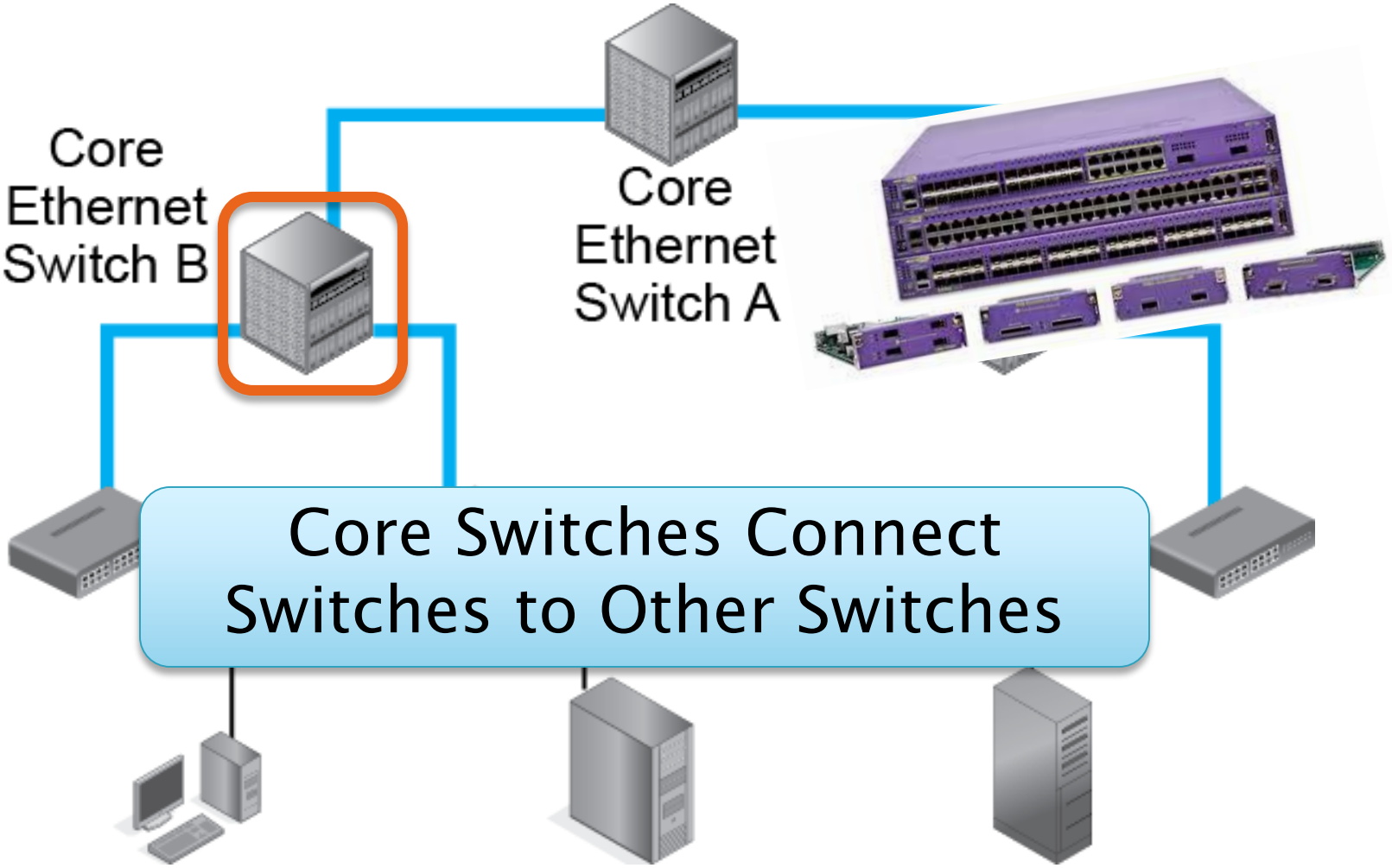


Server X

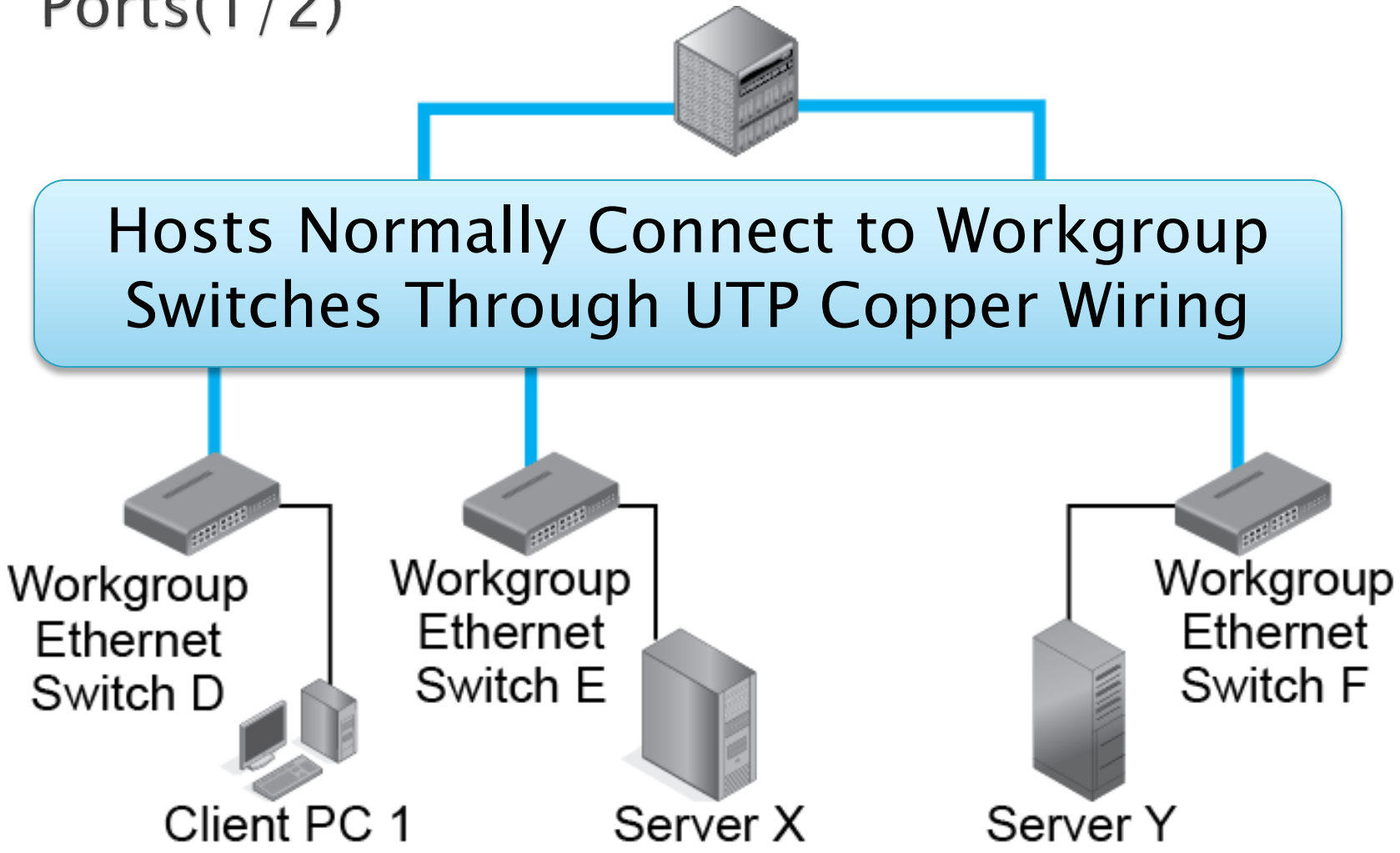


Server Y

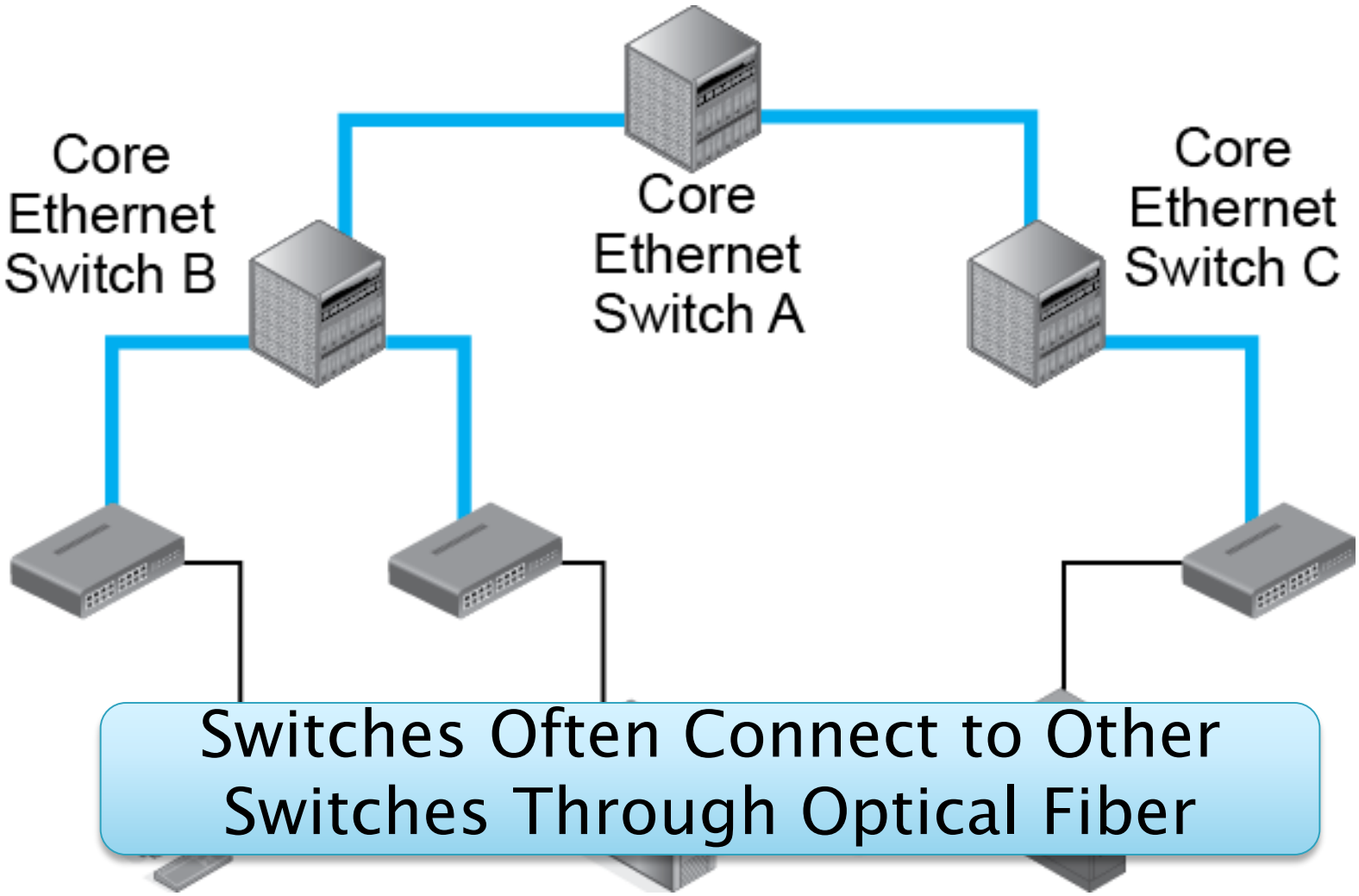
5.1 Simple Switched Ethernet Network(2/2)



5.2 Ethernet Workgroup Switch with 48 Ports(1/2)



5.2 Ethernet Workgroup Switch with 48 Ports(2/2)



Ethernet Begins

Introduction

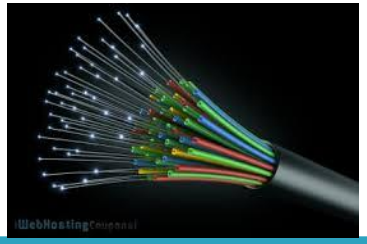
Physical Layer Standards

Data Link Layer Standards

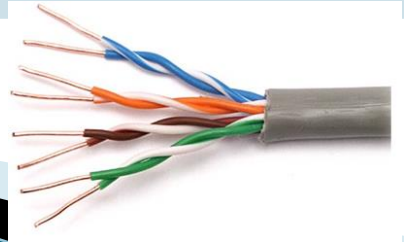
Advanced Switch Operation

Ethernet Security

5.3 UTP versus Optical Fiber



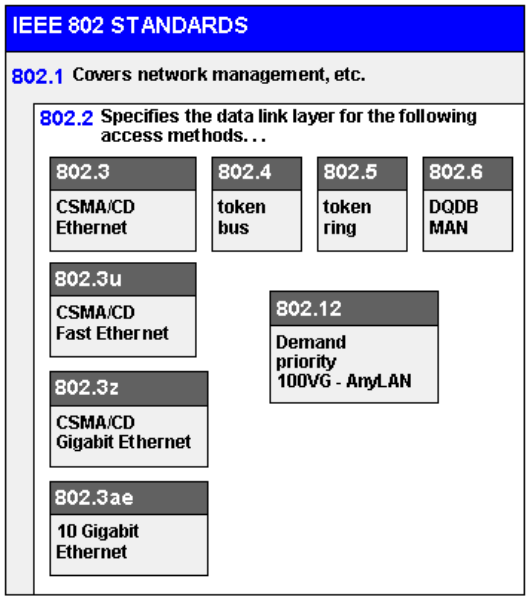
Characteristic	Unshielded Twisted Pair	Optical Fiber
Medium	Copper wire	Glass
Signal	Electrical	Light
Maximum Distance in LANs	Usually **100 m**	For LANs, usually **200 to 500 m**
Speed	Similar	Similar
Cost	Lower	Higher

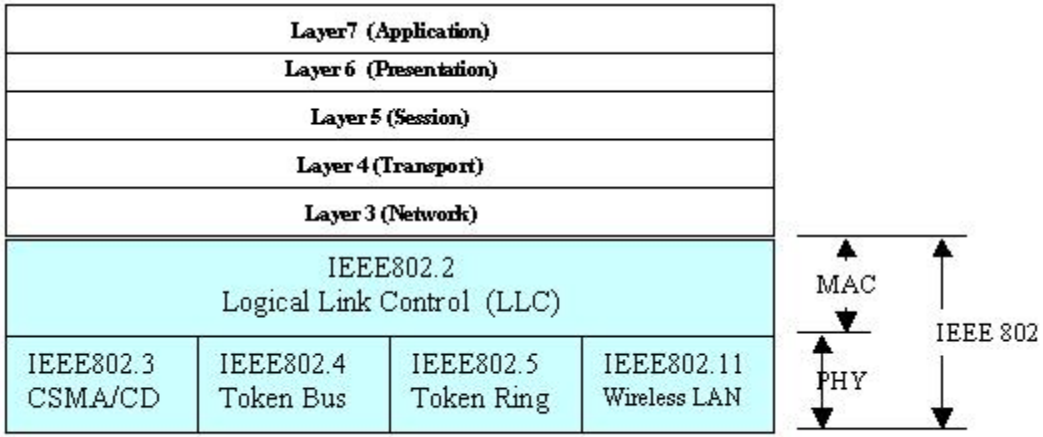


5.4 IEEE LAN/MAN Standards Committee

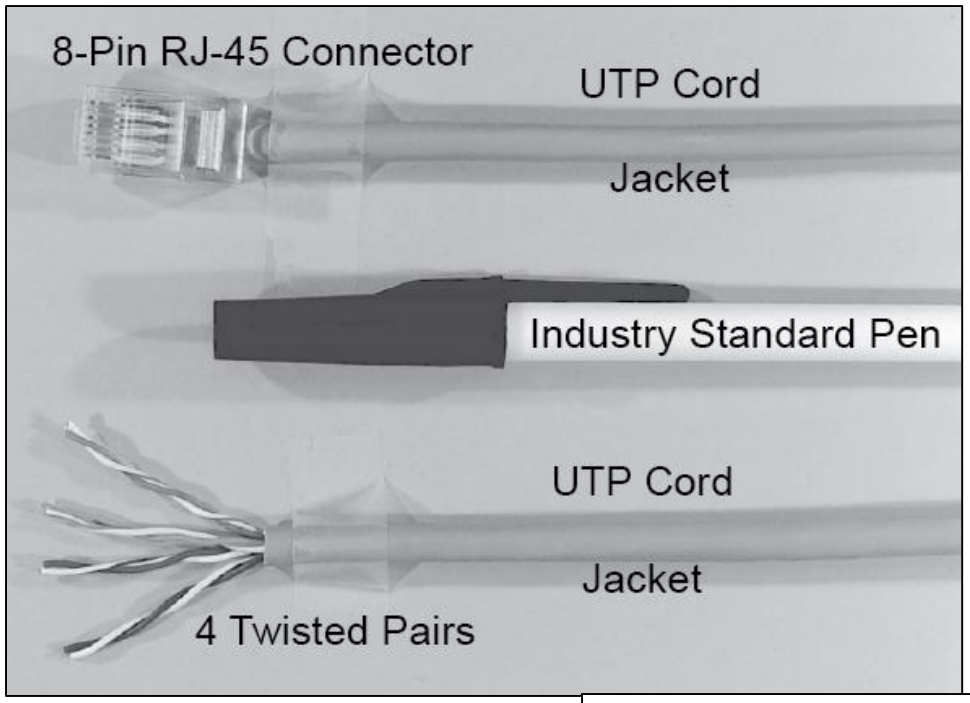
- IEEE 802 LAN/MAN Standards Committee
  - **802.1** : Standards used by multiple working groups, e.g. Security Standards
  - **802.3** : Ethernet standards
  - **802.11** : Wi-Fi wireless LAN standards

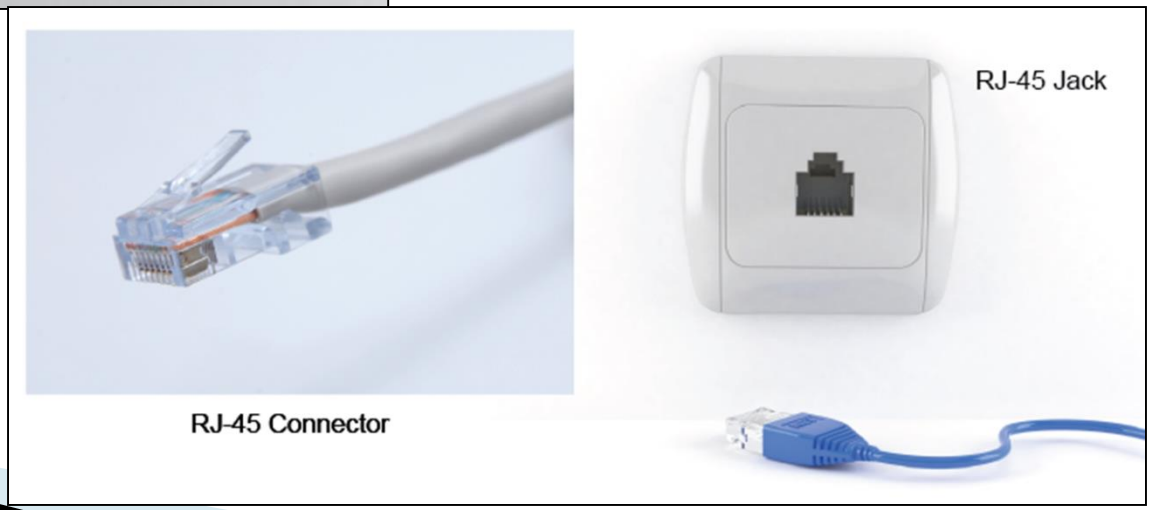
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5.8 Four-Pair Unshielded Twisted Pair Copper Wiring





5.9 RJ-45 (Ethernet) Connector and Jack



RJ-45 Connector



RJ-45 Jack

Q:你知道為何叫「RJ」嗎？

Registered Jack

5.11 UTP Quality Category, Transmission Speed, and Maximum Distance

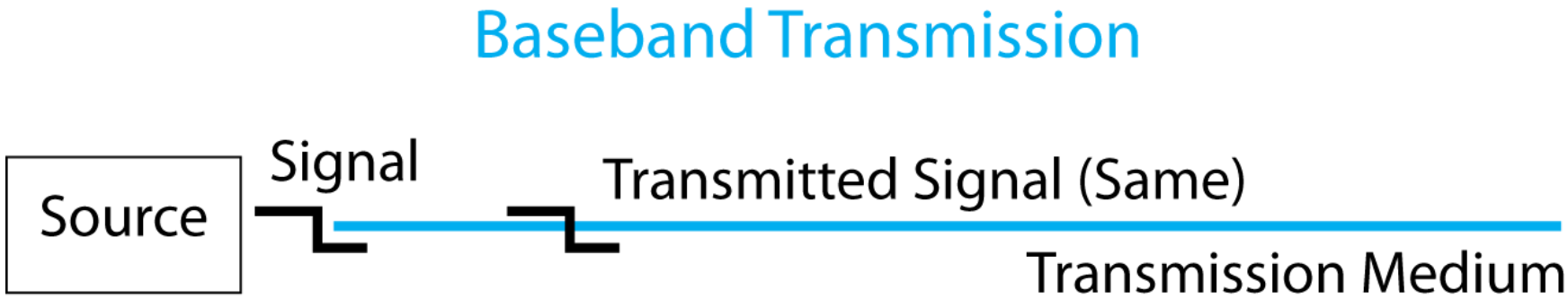
Ethernet Signaling Standard	Transmission Speed	UTP Quality Category	Maximum Cord Length
100BASE-TX	100 Mbps	Category 5e, 6, or higher	100 meters
1000BASE-T	1 Gbps	Category 5e, 6, or higher	100 meters
10GBASE-T	10 Gbps	Category 6	55 meters
10GBASE-T	10 Gbps	Category 6A	100 meters

Category is a measure of UTP QUALITY

Q:Do you know why 100**BASE**-XX?

小作業：請比較 Cat6, Cat7, Cat8之傳輸速率與應用之區別

Baseband v. Broadband Transmission (1/3)



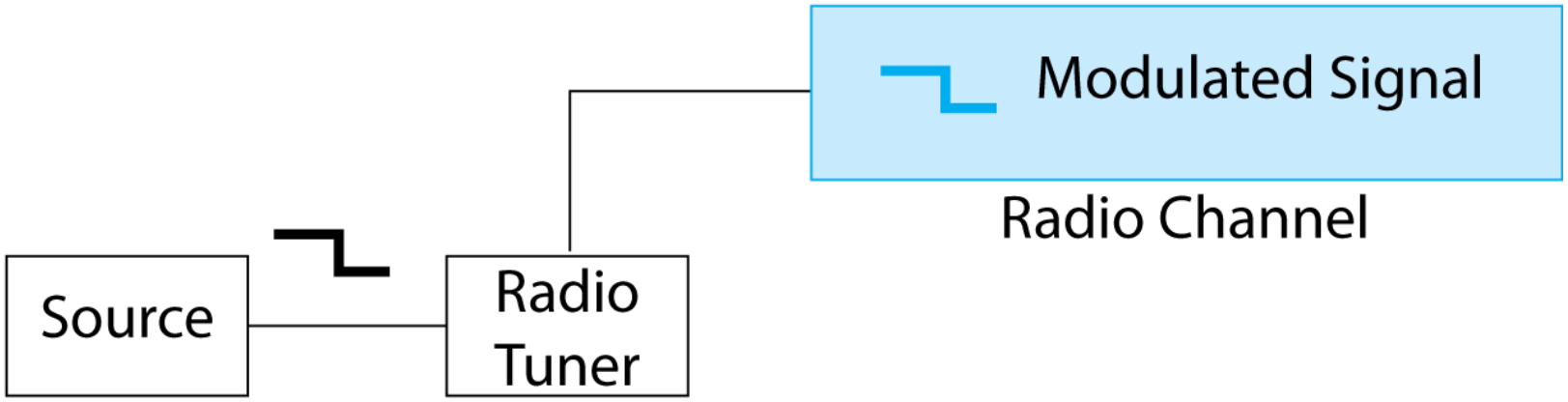
In baseband transmission, the signal is injected directly into the medium.

With UTP, change voltage levels.

With optical fiber, turn light on or off.

Baseband v. Broadband Transmission (2/3)

Broadband Transmission

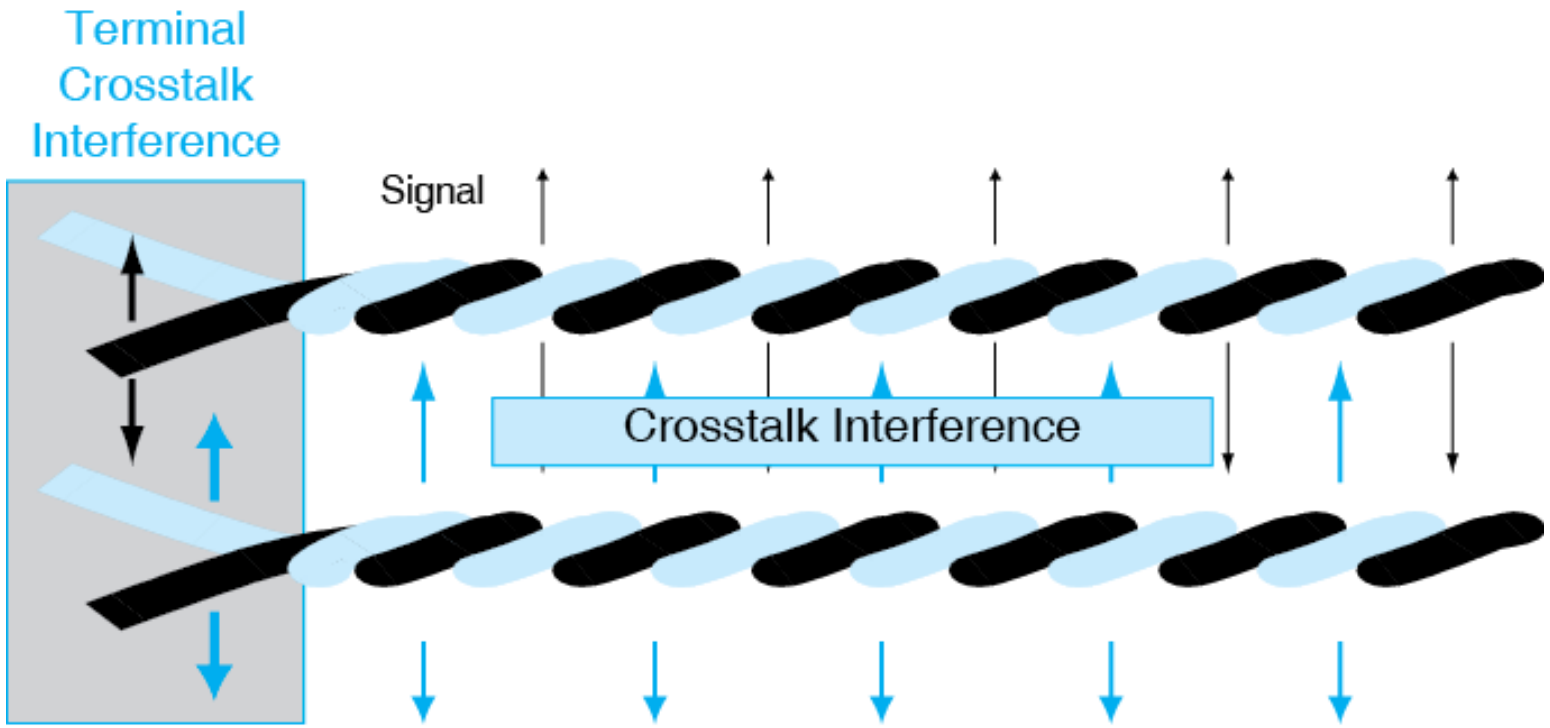


In broadband transmission, the signal is modulated into a radio channel and sent in a radio channel.

Baseband vs. Broadband Transmission (3/3)

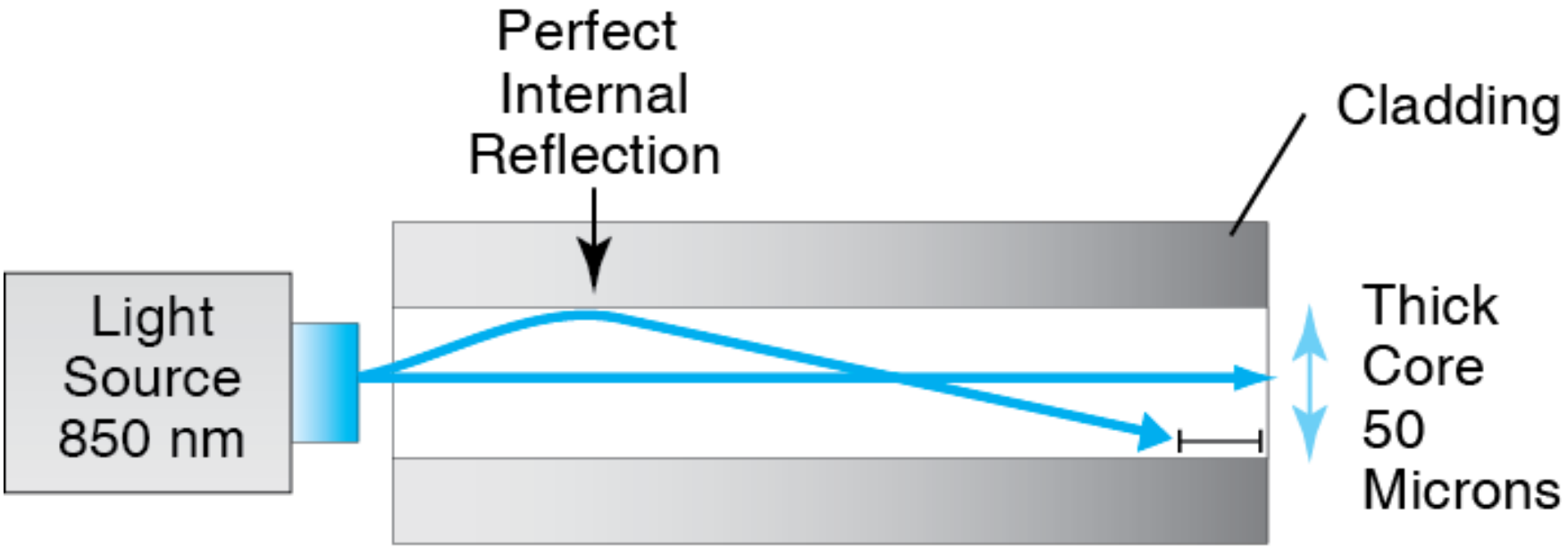
- Broadband transmission can carry signals farther by using amplifiers.
- However, baseband transmission is less expensive, and regeneration allows long distance transmission also.
- Baseband transmission is simpler, so it dominates LANs today.

5.12 Terminal Crosstalk Interference



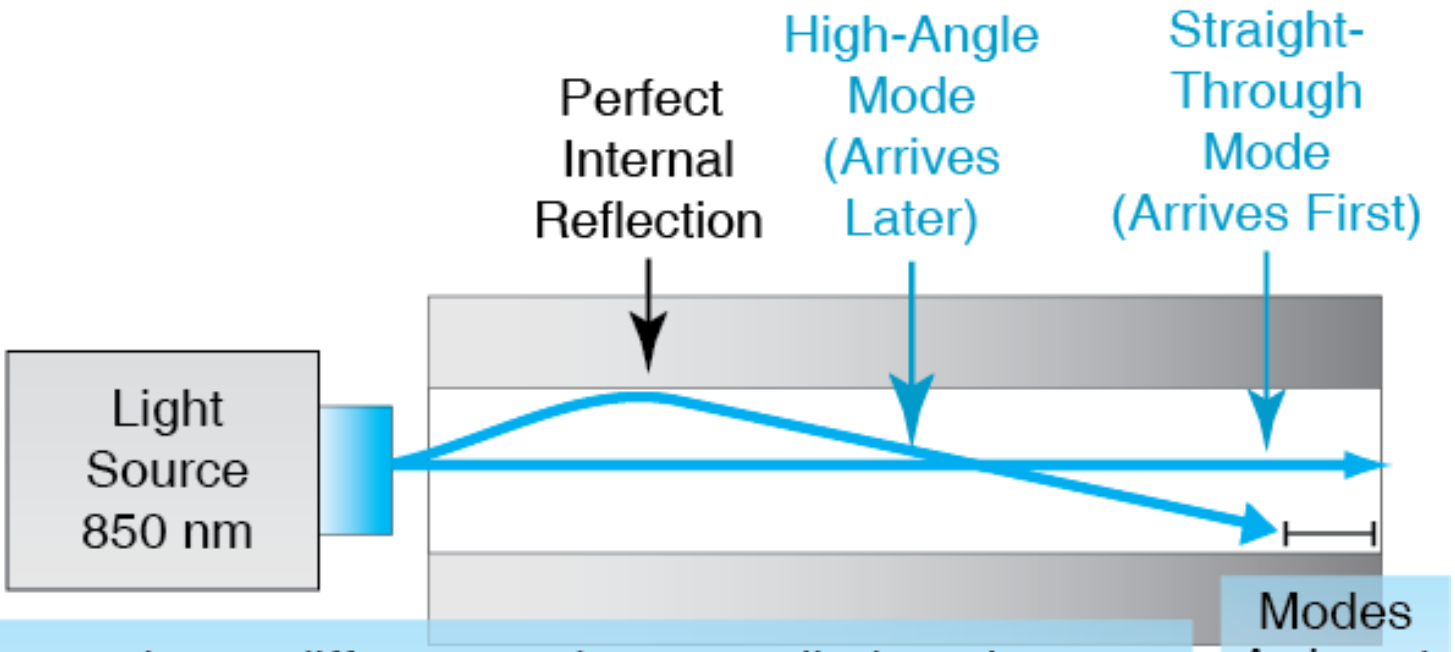
Crosstalk interference is worst at the ends, where the wires are untwisted to put them into the RJ-45 connector.
With no twisting, there is no protection from crosstalk interference.
This is terminal crosstalk interference.

5.13 Optical Fiber Transmission



When light rays hit the cladding, they are perfectly reflected into the core. There is no signal loss.

5.13 Optical Fiber Transmission



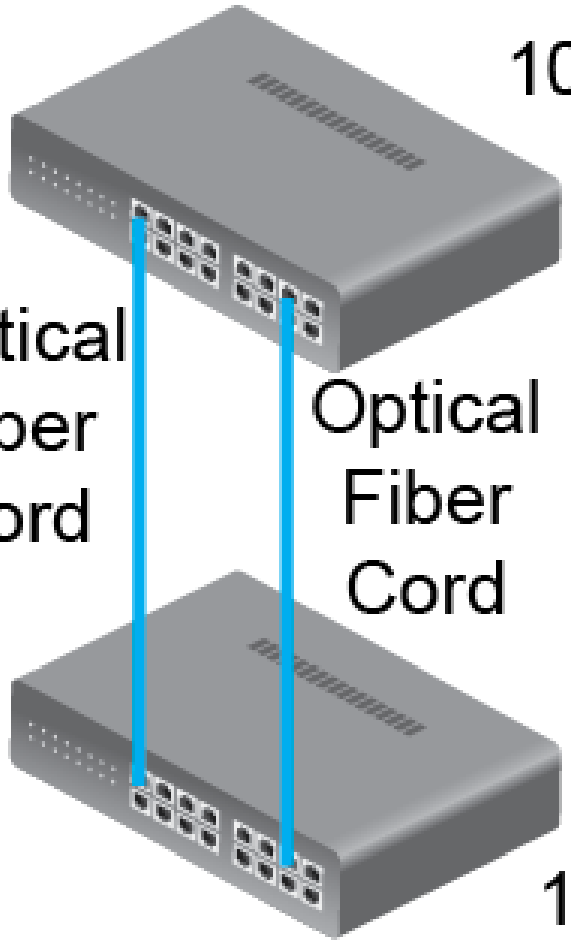
Light rays entering at different angles are called modes.
Light in different modes travels different distances.
This causes them to arrive at different times.
This is called modal dispersion.
If light from different clock cycles overlap, this will cause errors.

5.16 Optical Fiber Quality Designations and Transmission Distance

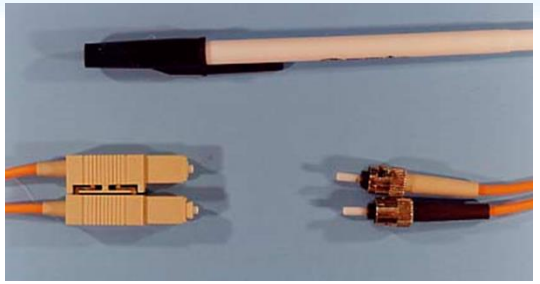
ISO 11801 Standard	Minimum Modal Bandwidth (MHz*km) at 850 nm	Maximum Distance at 1 Gbps, 850 nm light (1000BASE-SX)	Maximum Distance at 10 Gbp, 850 nm light (10GBASE-SR)	Maximum Distance at 100 Gbps, 850 nm light
OM3	2,000	550 m	300 m	100 m
OM4	4,700	1,000 m	440 m	150 m

OM stands for **Optical Multimode**; a standard for **multimode fiber.**

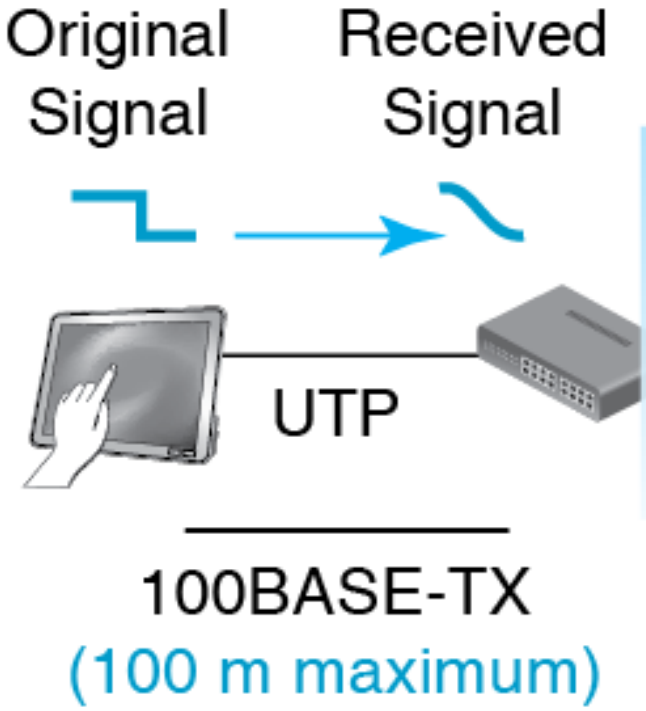
5.17 Link Aggregation (Bonding)



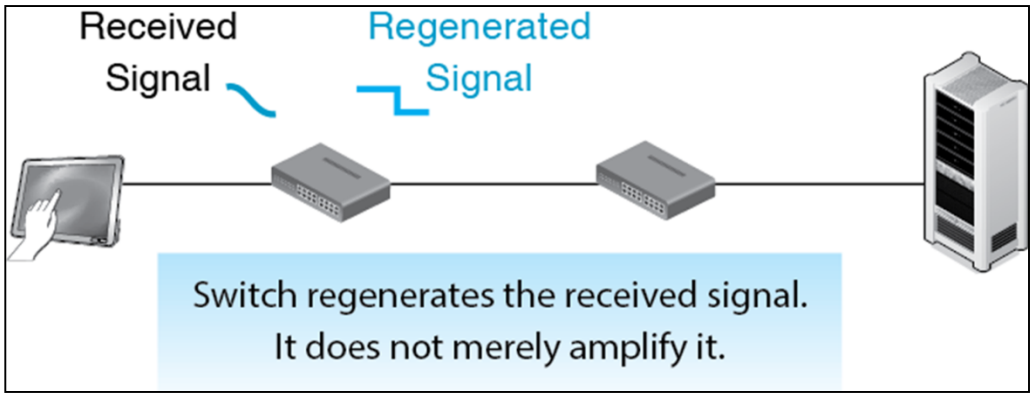
Two links provide 2 Gbps of trunk capacity between the switches.



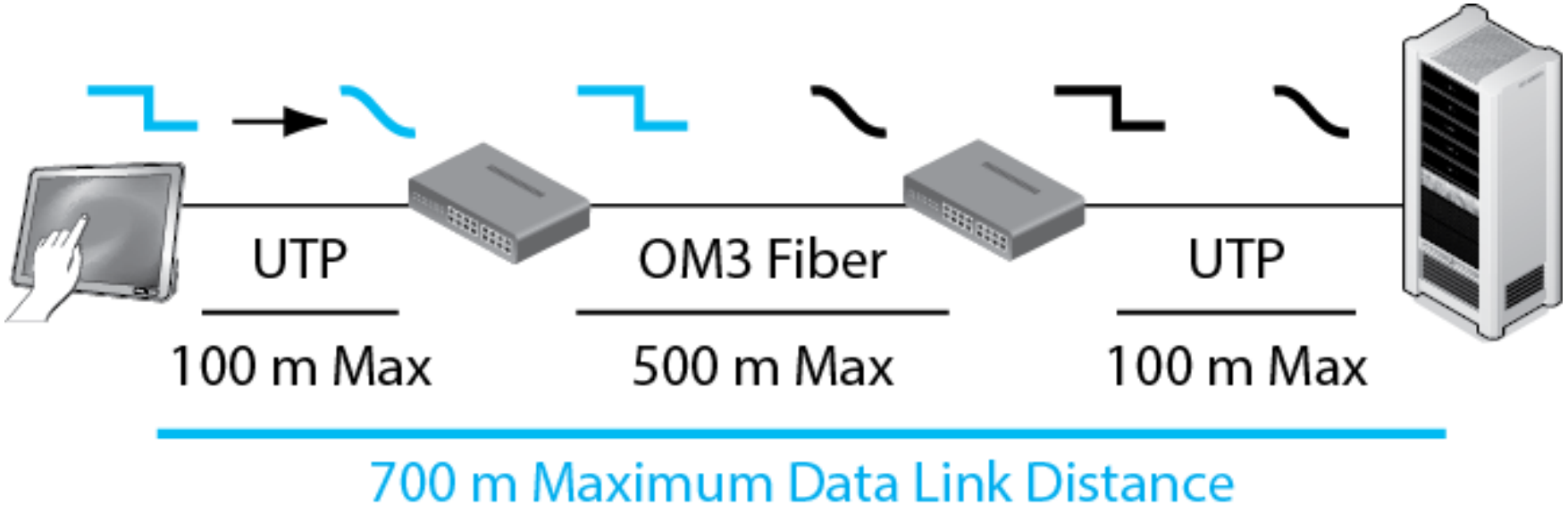
5.18 Ethernet Physical Link Maximums and Unlimited Data Link Distances



UTP has a maximum transmission distance of 100 m. Longer distance will degrade the signal too much to read correctly



5.18 Ethernet Physical Link Maximums and Unlimited Data Link Distances



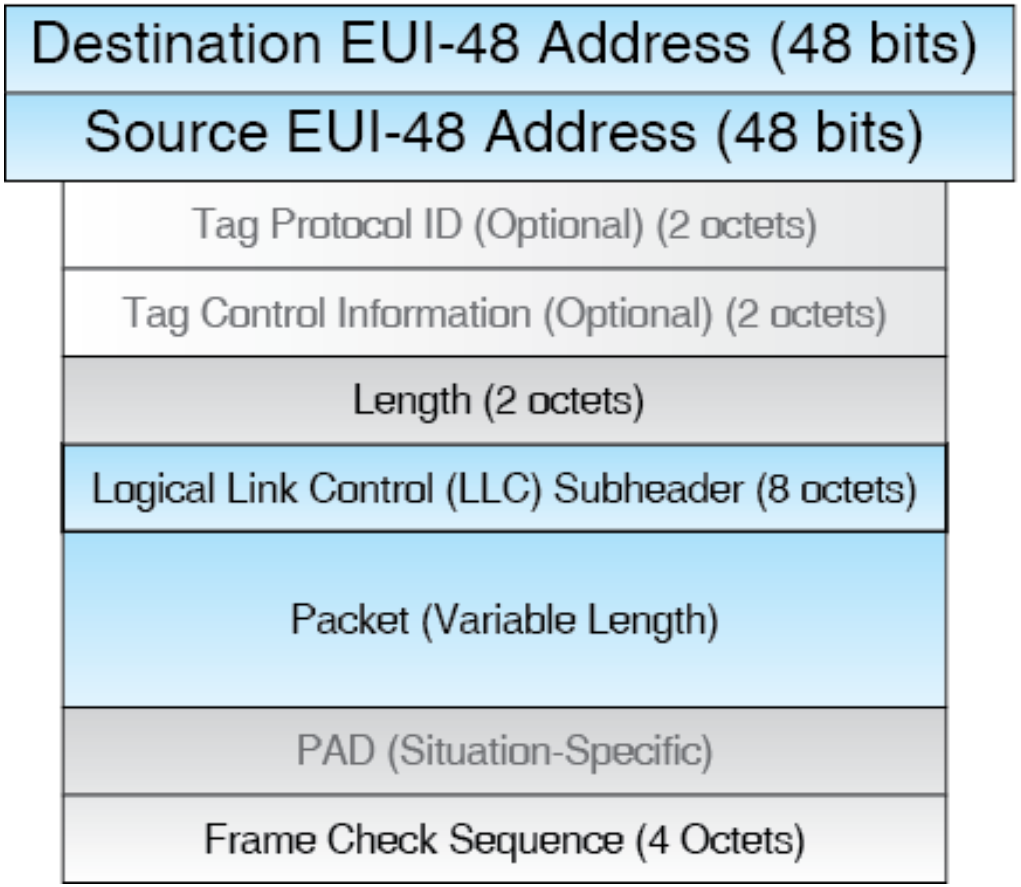
Regeneration can happen multiple times.
Physical link distance limits do not apply to data links.

- Ethernet Begins
- Introduction
- Physical Layer Standards
- **Data Link Layer Standards**
- Advanced Switch Operation
- Ethernet Security

Creating Ethernet Standards

- IEEE 802.3 for Ethernet standards
- IEEE 802.11 for wireless LAN standards
- IEEE 802.16 for WiMax wireless metropolitan area network standards
- The terms *802.3* and *Ethernet* are interchangeable today.
- Ethernet **also** has data link layer standards (frame organization, switch operation, etc.).

5.19 Ethernet 802.3 Frame



For humans, EUI-48 addresses are expressed in hexadecimal notation

Ethernet MAC Layer Frame

Field
Preamble (7 octets of 10101010 for synchronization)
Start Frame Delimiter (10101011 to end synch)

The first two fields synchronize the receiver's clock with the senders clock.

If this was not done, the receiver might read bit $1{,}01_2$ when it is really bit $1{,}10_2$.

Ethernet MAC Layer Frame

Field
Preamble (7 octets of 10101010 for synchronization
Start Frame Delimiter (10101011 to end synch)
Destination MAC address (48 bits)
Source MAC address (48 bits)
Tag Protocol ID (TPID, Optional, 2 octets)
Tag Control Information (TCI, optional, 2 octets)
Length (2 octets)
PAD (Situation-Specific)
Frame Check Sequence

The length field gives the length of the data field, not the total length of the frame.

Ethernet MAC Layer Frame

Field
Packet (variable length)
PAD (Situation-Specific)
Frame Check Sequence

The PAD field is added by the sender only if the data field is less than 46 octets.

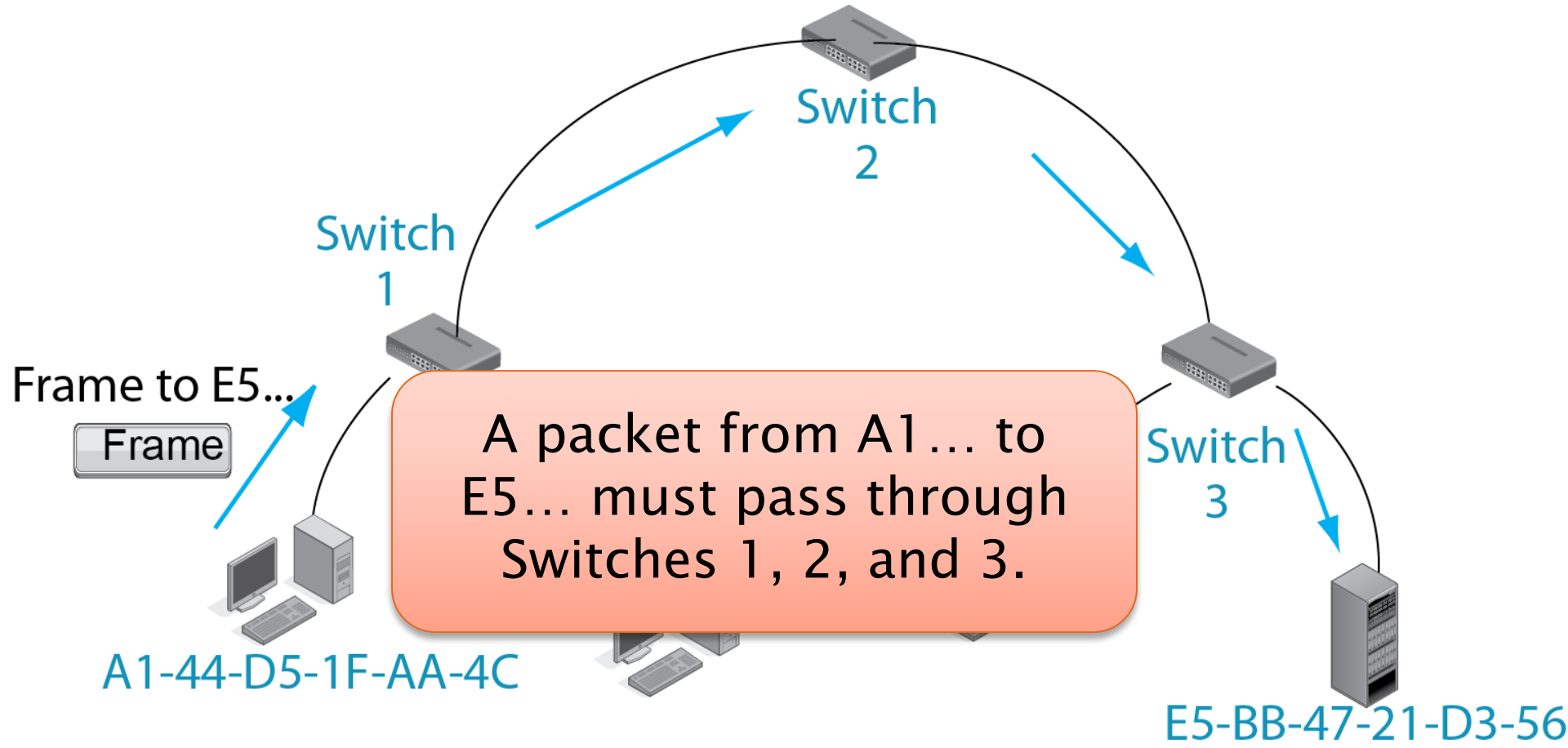
The PAD field is selected so that the total of the length field and the pad is 46 octets.

Ethernet MAC Layer Frame

Field
Preamble (7 octets of 10101010 for synchronization
Start Frame Delimiter (10101011 to end synch)
Packet (variable length)
PAD (Situation-Specific)
Frame Check Sequence

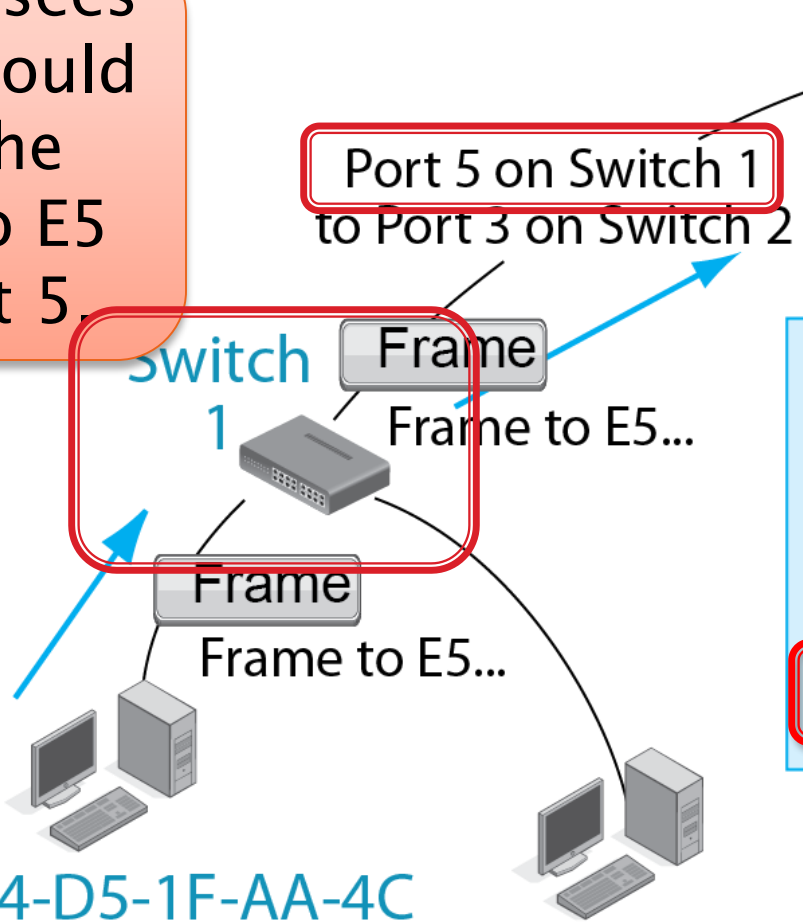
The Frame Check Sequence field is for error detection.
If an error is found, the frame is discarded.
There is no error message or request for transmission.
Ethernet is not reliable.

5.21 Multi-Switch Ethernet Operation



5.21 Multi-Switch Ethernet Operation

Switch 1 sees that it should send the frame to E5 out Port 5.

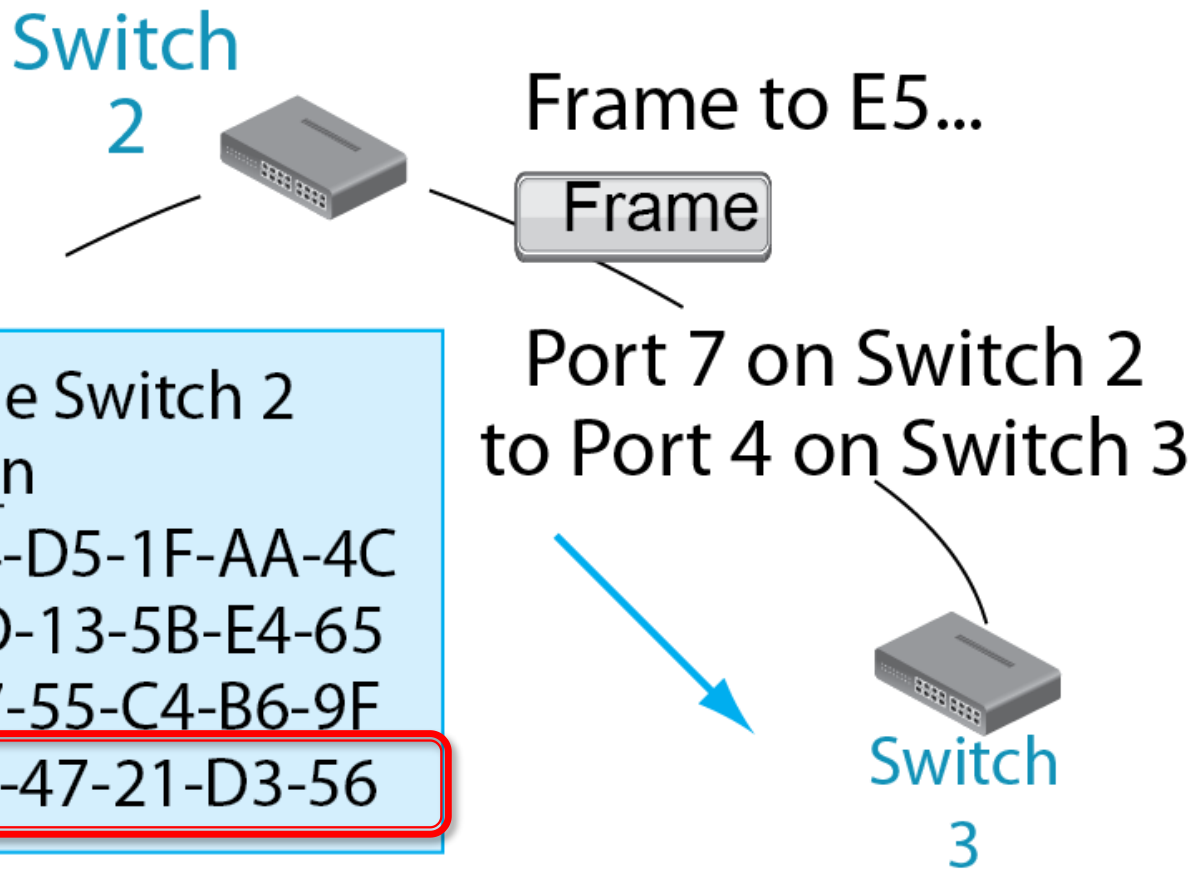


Switching Table Switch 1

Port	Station
2	A1-44-D5-1F-AA-4C
7	B2-CD-13-5B-E4-65
5	D4-47-55-C4-B6-9F
5	E5-BB-47-21-D3-56

5.21 Multi-Switch Ethernet Operation

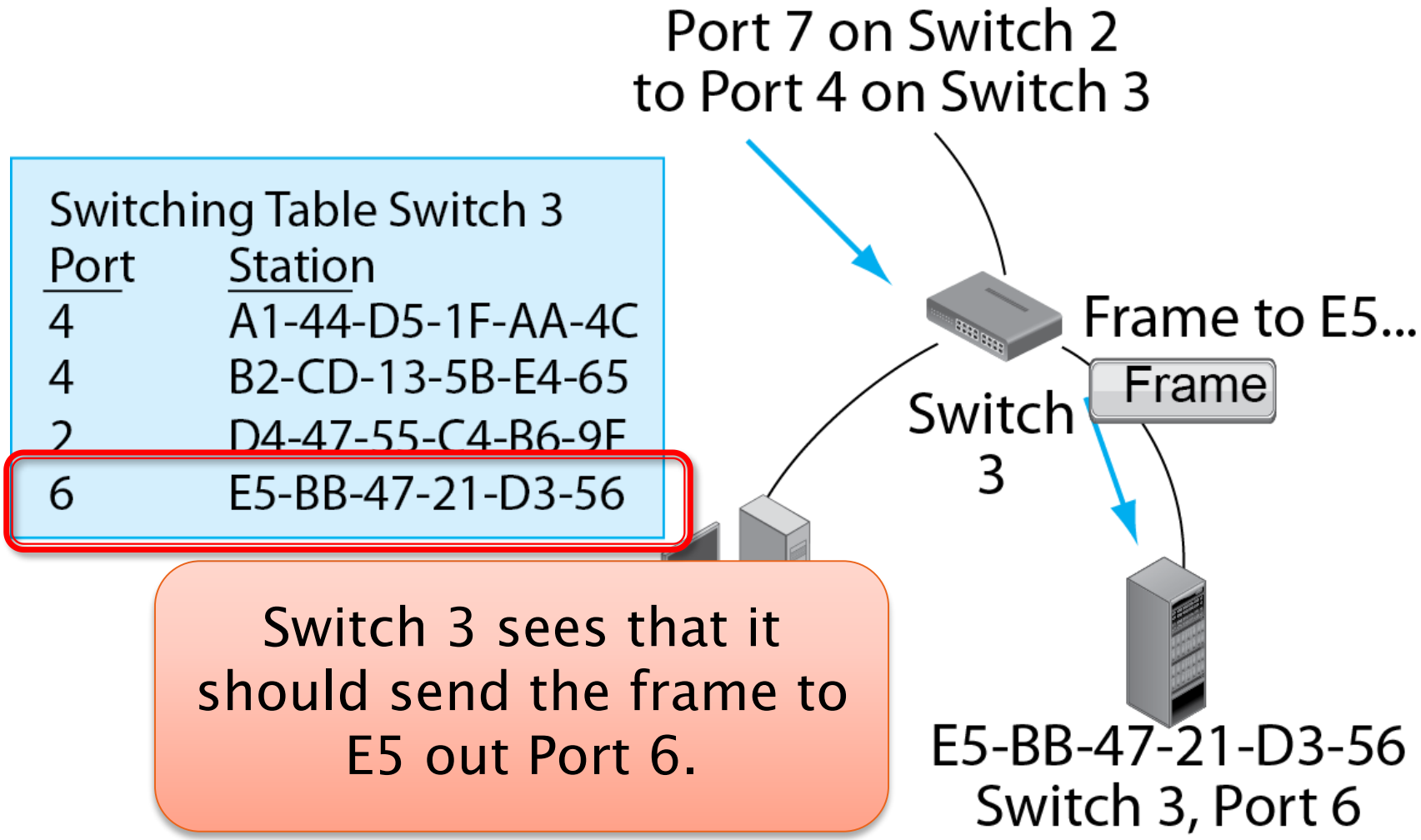
Switch 2 sees that it should send the frame to E5 out Port 7.



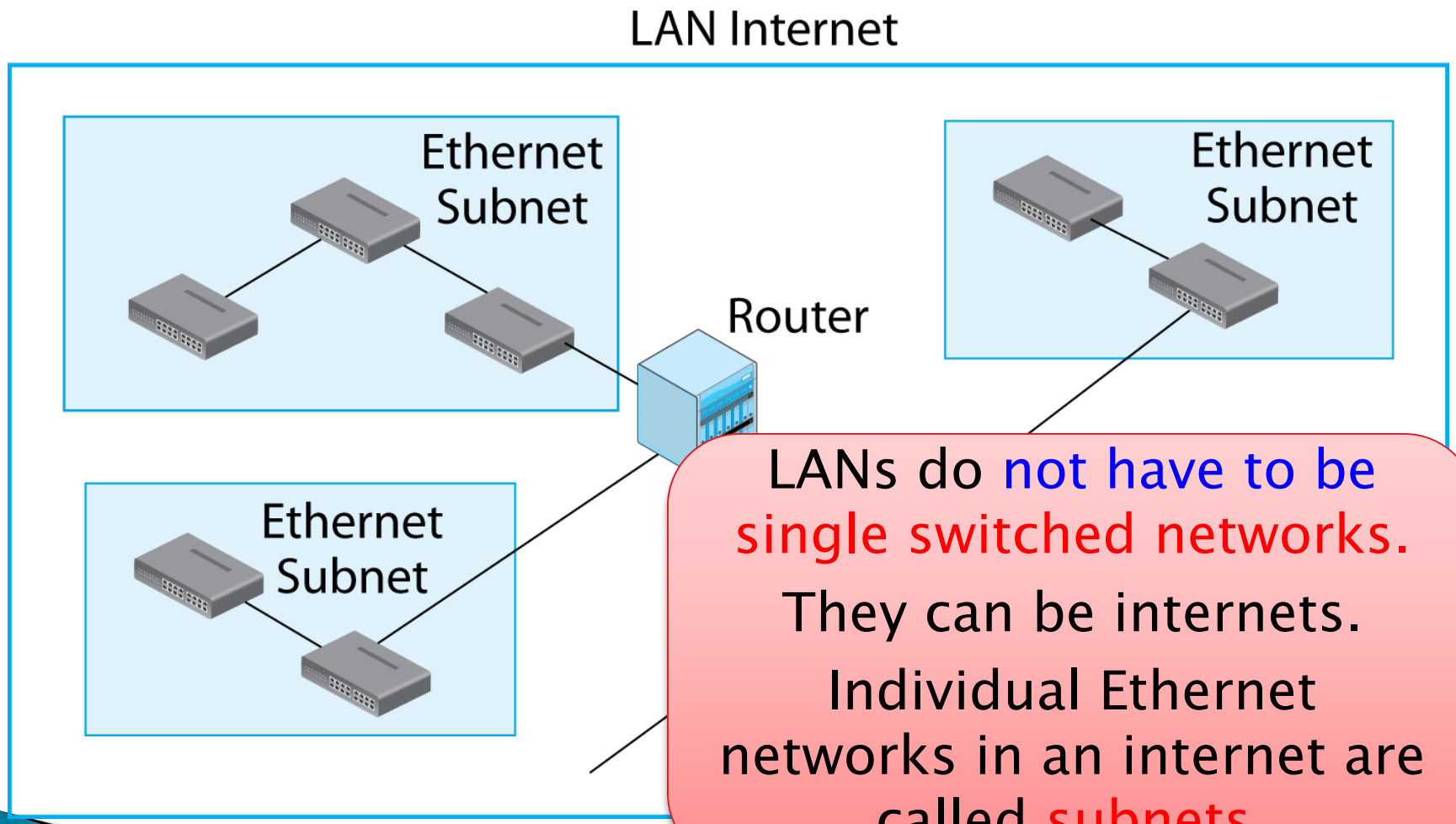
Switching Table Switch 2

Port	Station
3	A1-44-D5-1F-AA-4C
3	B2-CD-13-5B-E4-65
7	D4-47-55-C4-B6-9F
7	E5-BB-47-21-D3-56

5.21 Multi-Switch Ethernet Operation



LAN Internet with Ethernet Subnets



LANs do not have to be single switched networks.

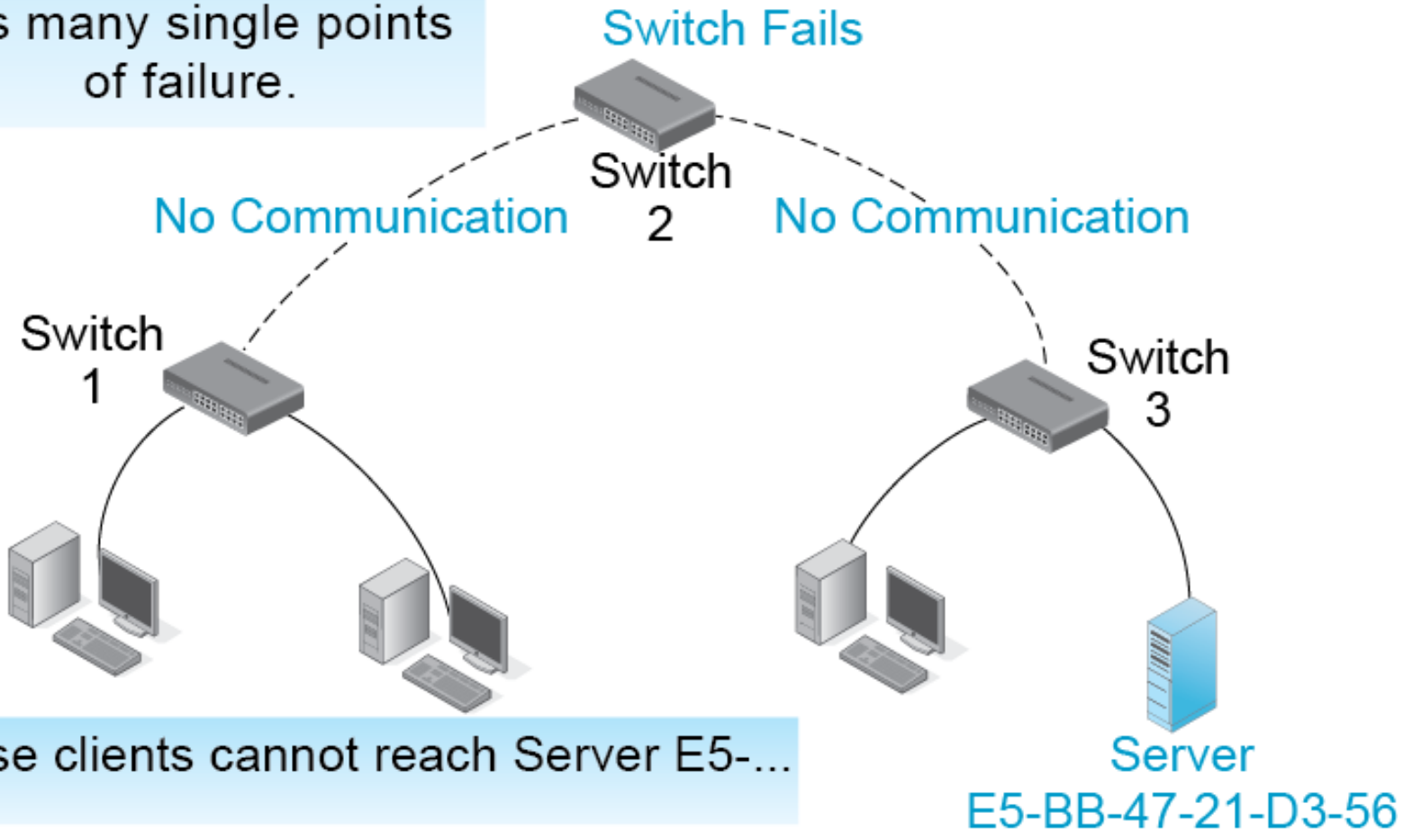
They can be internets.

Individual Ethernet networks in an internet are called subnets.

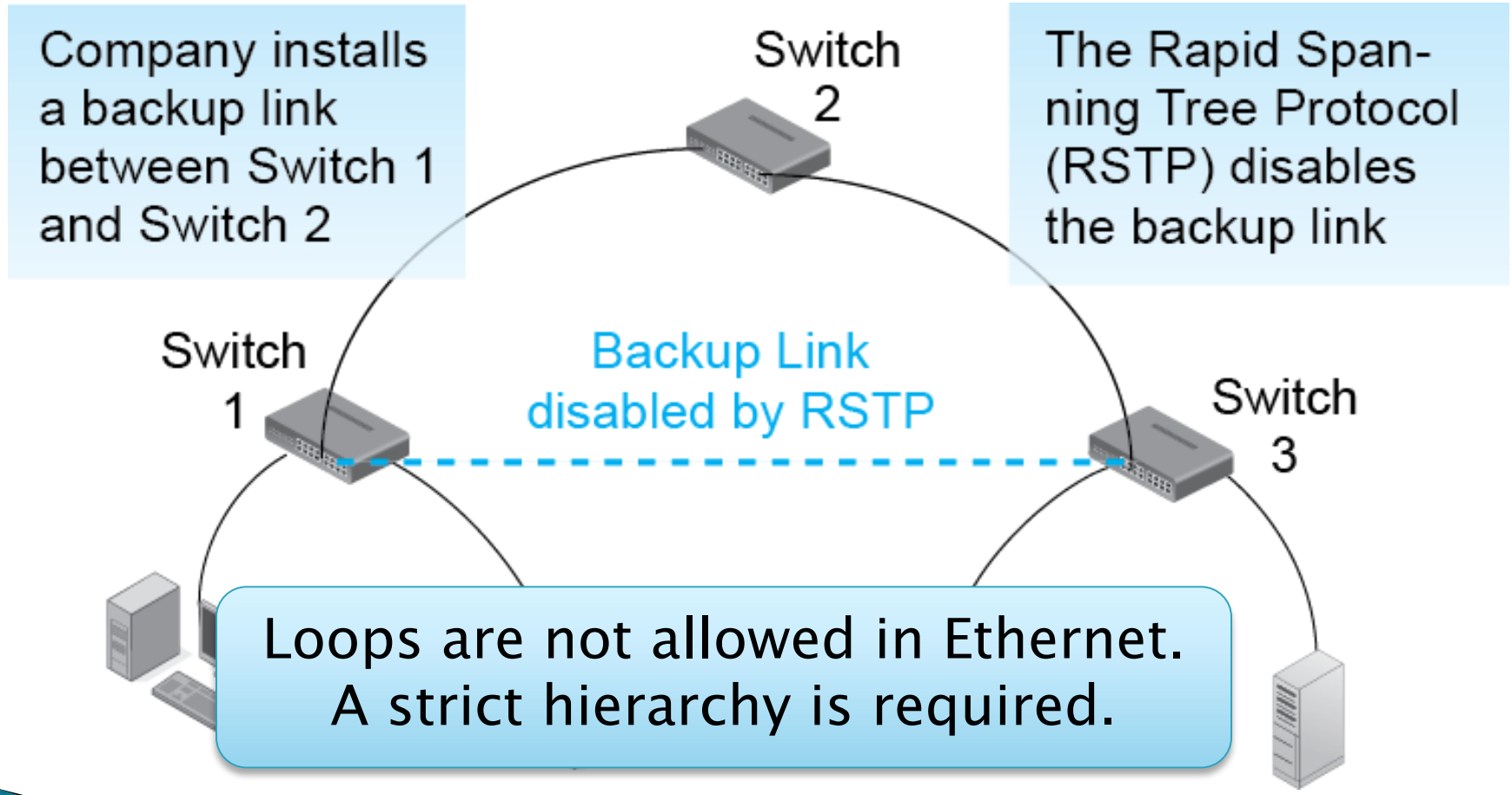
- Ethernet Begins
- Introduction
- Physical Layer Standards
- Data Link Layer Standards
- **Advanced Switch Operation**
- Ethernet Security

5.22 Single Point of Failure

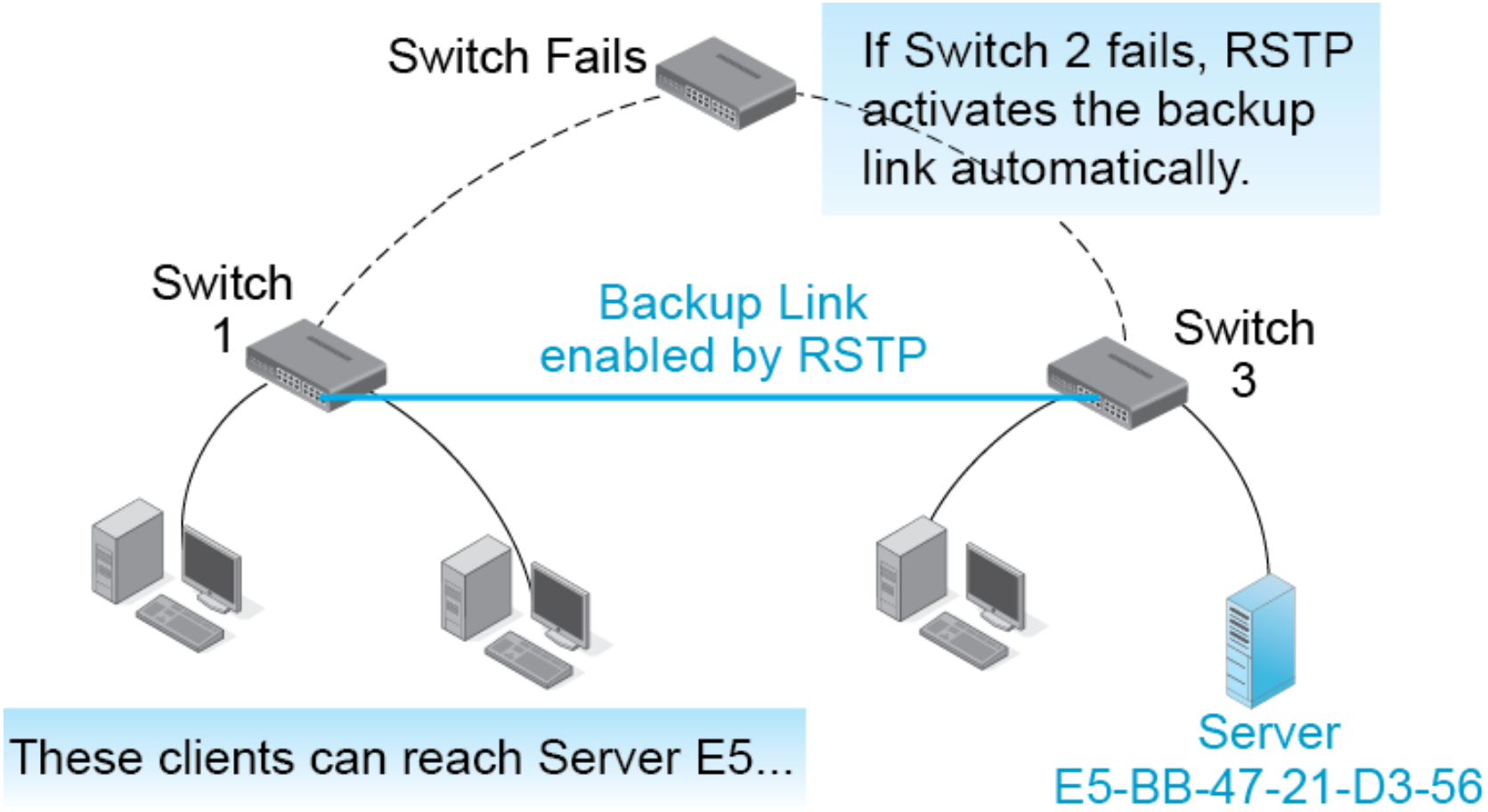
A hiearchical network has many single points of failure.



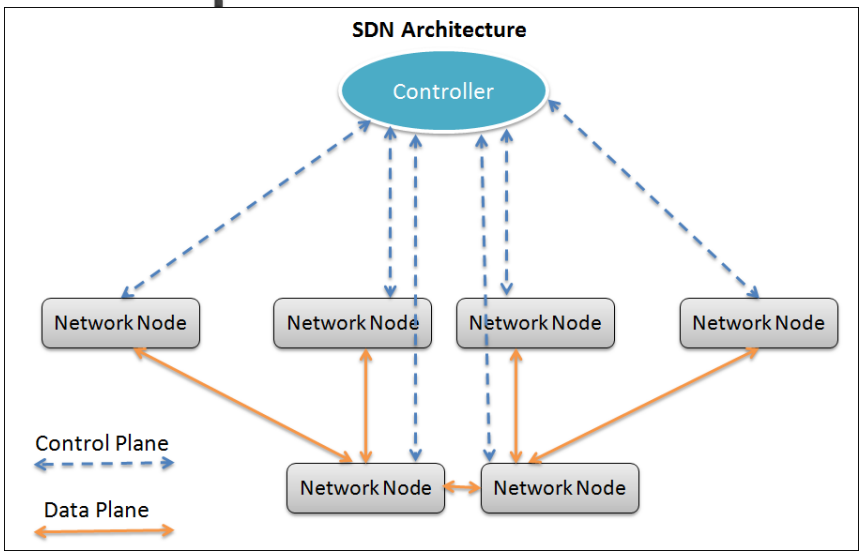
5.23 Backup Link and the Rapid Spanning Tree Protocol



5.23 Backup Link and the Rapid Spanning Tree Protocol (Cont.)



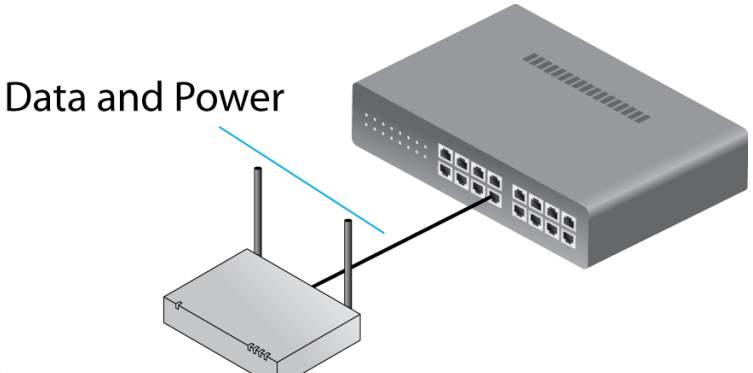
5.24 Advanced Ethernet Capabilities



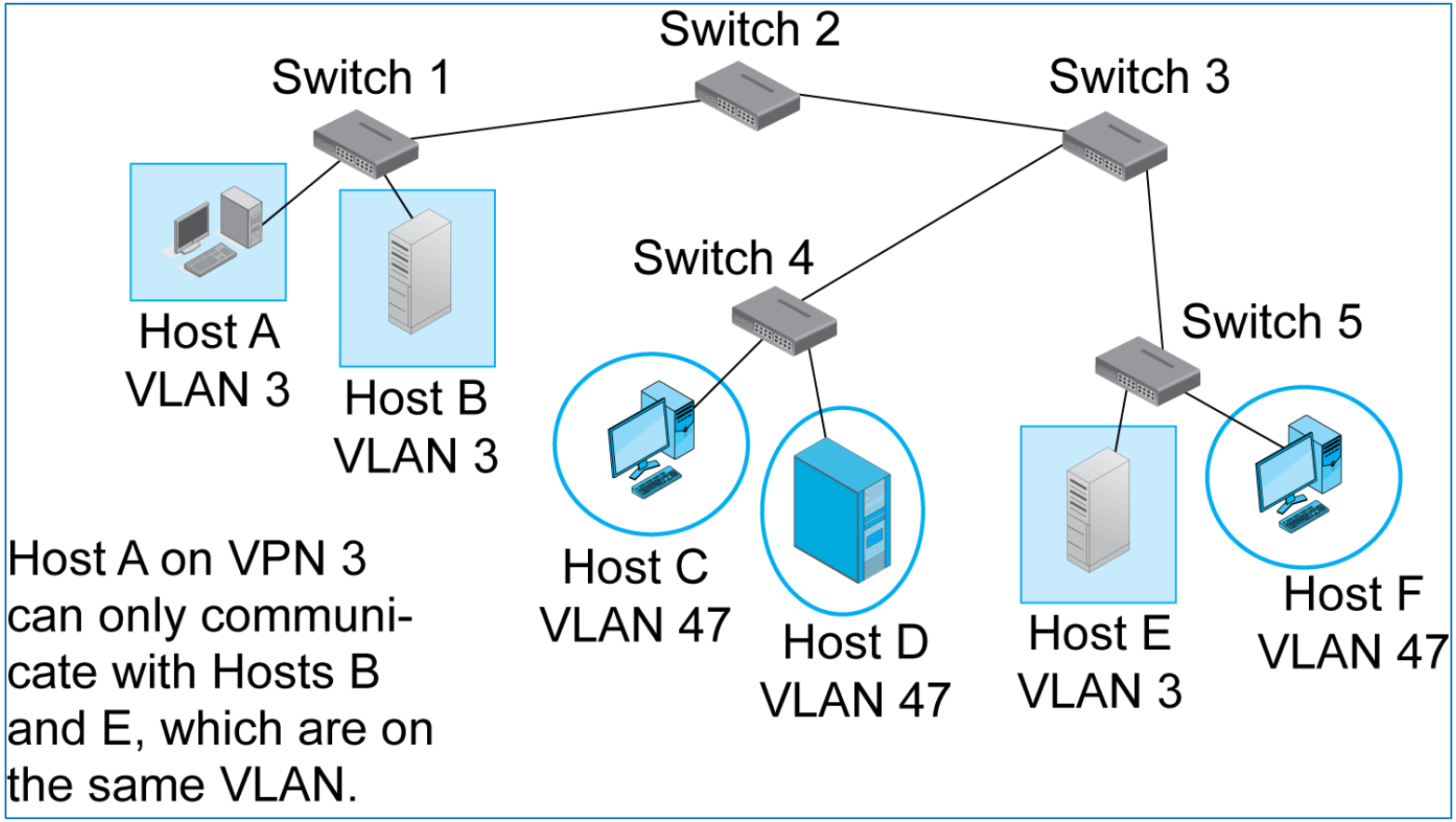
- **Priority**
  - up to eight priority levels
- **Manageability**
  - managed by SNMP
  - Although cost much more, it is made up for lower management costs
  - **Software-defined networking** may bring a revolution in switch management

5.24 Advanced Ethernet Capabilities (Cont.)

- **Power over Ethernet (POE)**
  - Switches can supply power to devices **via UTP**. (Wired telephone systems and USB ports already do this.)
  - Switches implementing POE Plus can provide up to **25 watts** of power over UTP to hosts, sufficient for voice over IP phones, surveillance cameras, and most access points



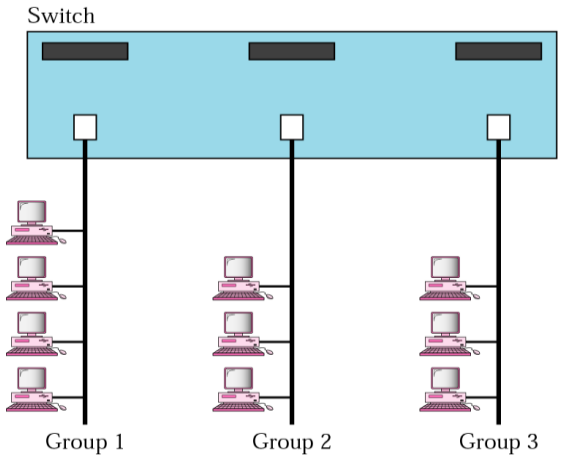
5.24 Advanced Ethernet Capabilities (Cont.) –Ethernet Virtual LANs (VLANs)

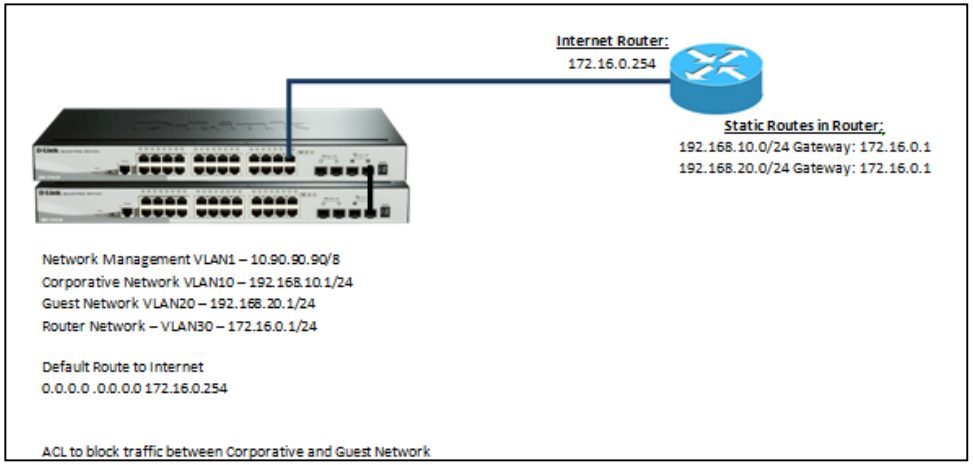


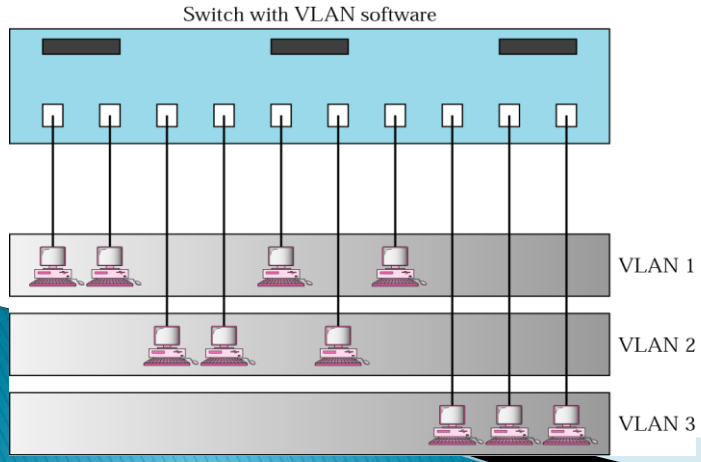
+ VLANs reduce traffic when hosts broadcast.
+VLANs give security.

Virtual LANs

- A LAN configured by software, not by hardware







Traditional Ethernet data frame

6 bytes	6 bytes	2 bytes	46-1500 bytes	4 bytes
Destination address	Source address	Length/Type	Data	FCS

VLAN data frame

6 bytes	6 bytes	4 bytes	2 bytes	46-1500 bytes	4 bytes
Destination address	Source address	VLAN Tag	Length/ Type	Data	FCS

TPID	PRI	CFI	VID
2 bytes	3 bits	1 bit	12 bits

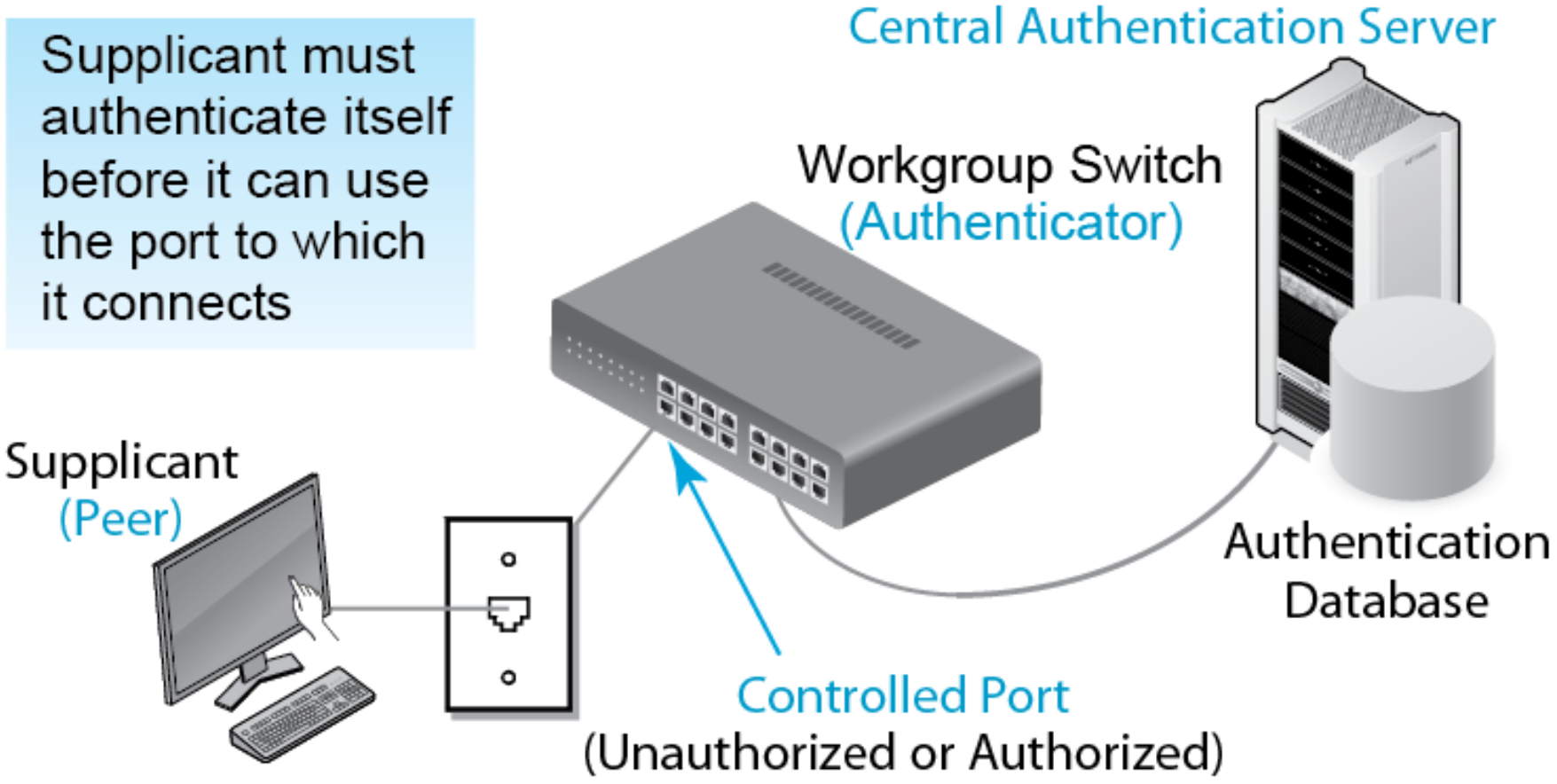
- Ethernet Begins
- Introduction
- Physical Layer Standards
- Data Link Layer Standards
- Advanced Switch Operation
- **Ethernet Security**

5.25 802.1X Port-Based Access Control on an Ethernet Switch



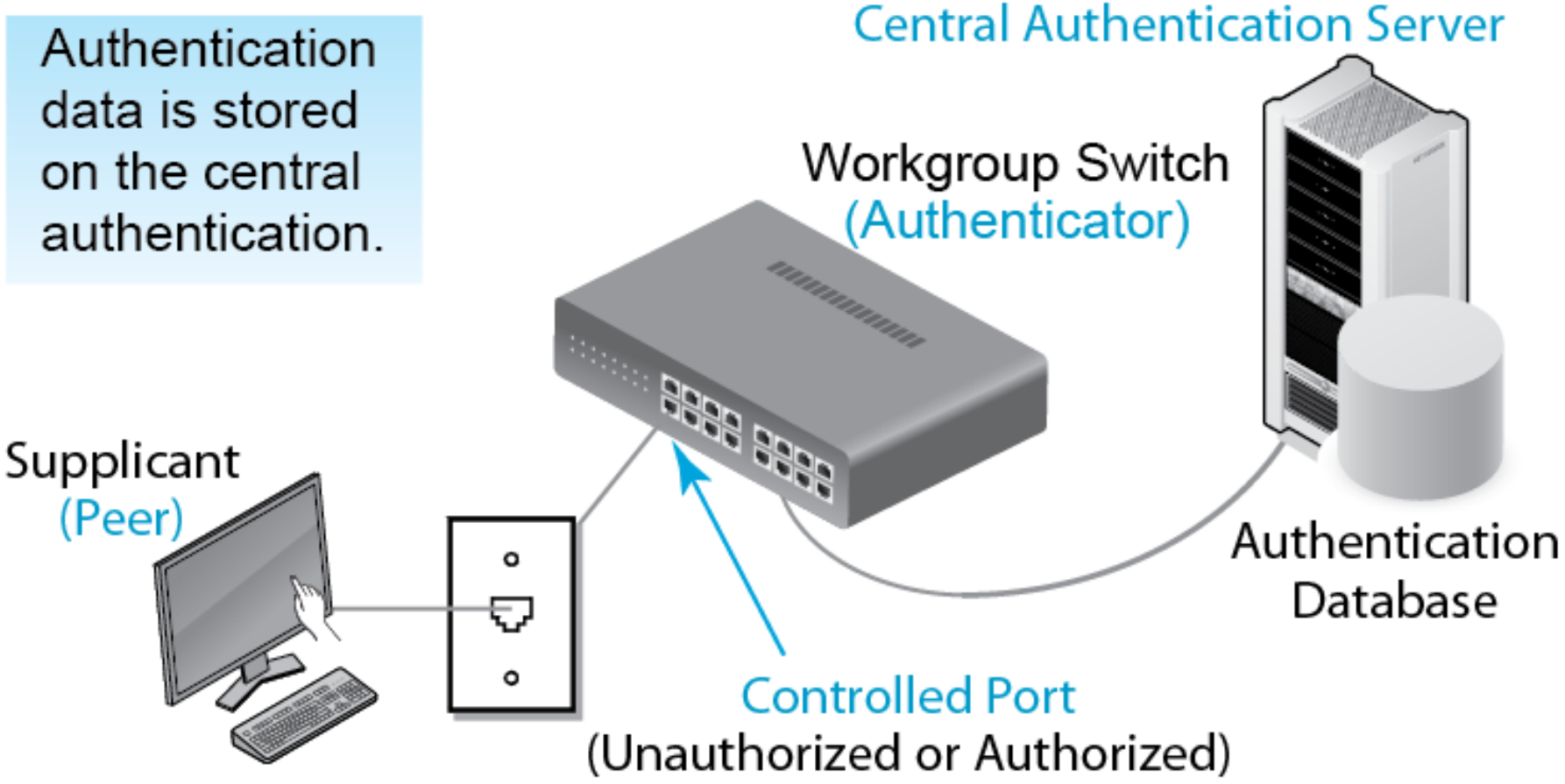
- **The Problem**
  - **Anyone** can enter the building and plug their computer into a switch or into a wall **RJ-45 port**,.
  - This usually gives the attacker access to the network <u>without going through a firewall</u>.
- **Solution: access control at switch ports.**
  - **802.1X Port Based Access Control**. Not created by the 802.3 WG, but the **802.1 WG**, which creates general standards, such as security standards.
- WG: Working Group

5.25 802.1X Port-Based Access Control on an Ethernet Switch (1/4)



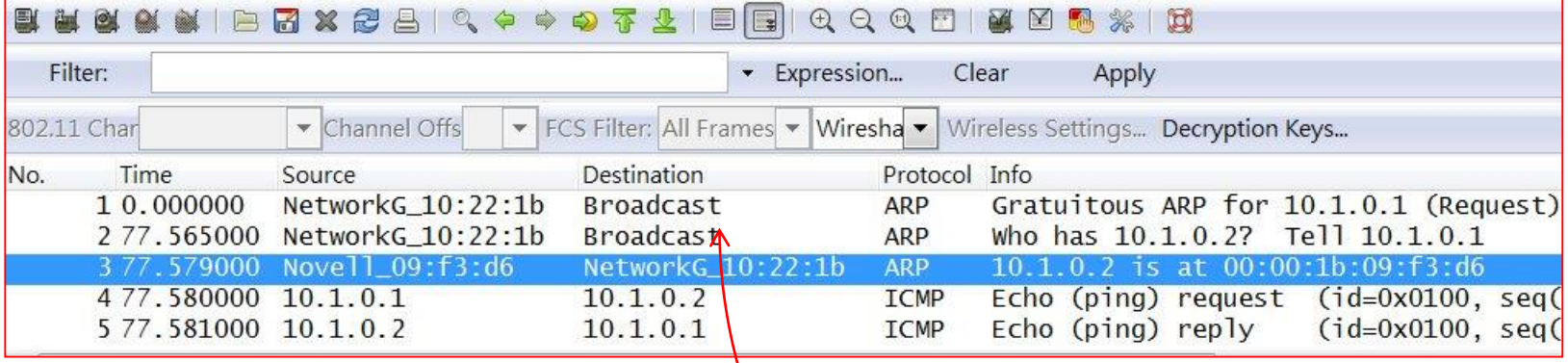
5.25 802.1X Port-Based Access Control on an Ethernet Switch (2/4)

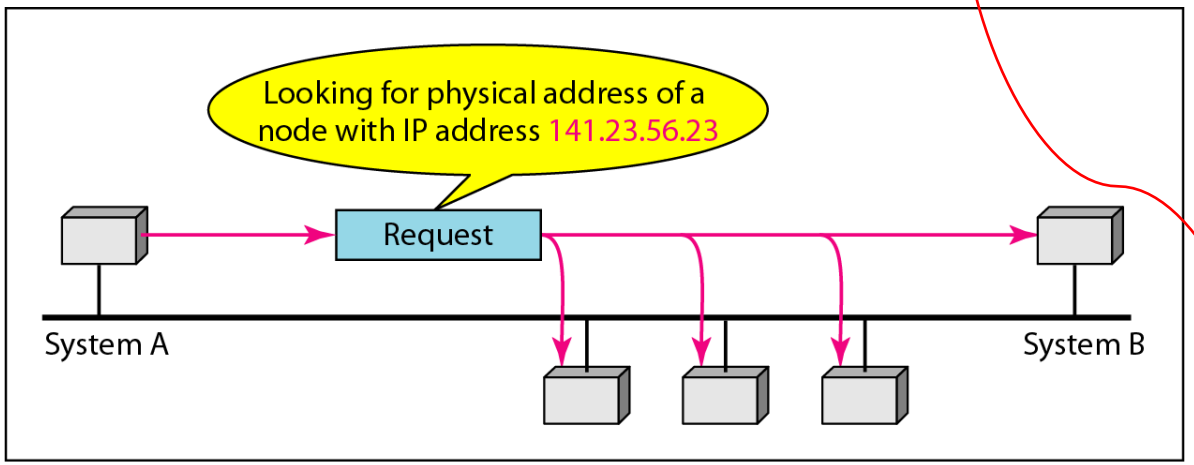
Authentication data is stored on the central authentication.



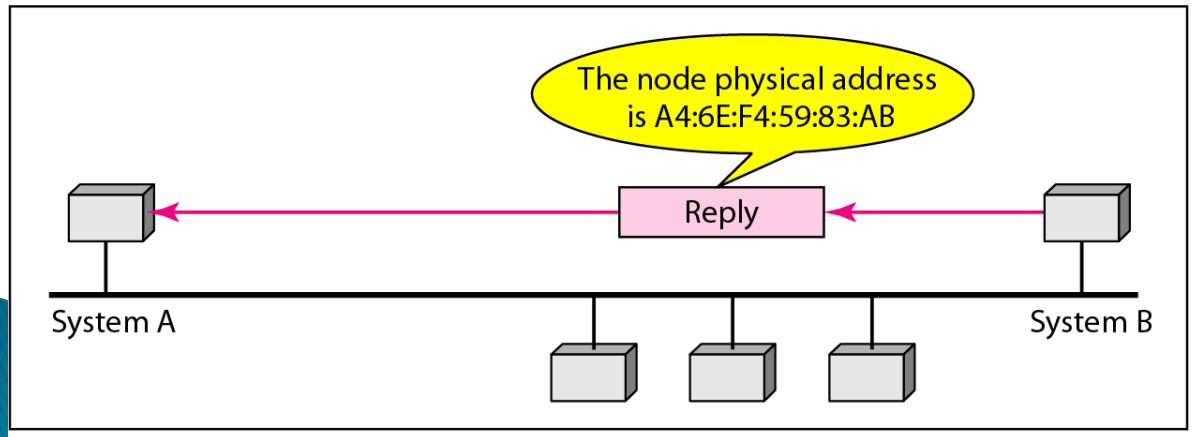
5.25 802.1X Port-Based Access Control on an Ethernet Switch (3/4)

- Advantages of a **Central Authentication Server**
  - **Consistency**: Attacker cannot find a misconfigured switch.
  - **Rapid changes**: When someone leaves, is hired, or needs credential changes.
  - **Switch cost**: Authentication server does heavy work.
  - **Reduced management cost**: Only one authentication database to maintain.





a. ARP request is broadcast



b. ARP reply is unicast

ARP（Address Resolution Protocol）原理

An ARP request is **broadcast**; an ARP reply is unicast.

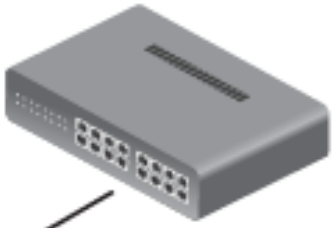
5.26 Man in the Middle Attack in an Ethernet LAN Using ARP Poisoning (1/4)



Attacker

EUI-48 = FF-FF-FF-DD-DD-DD

ARP Cache tells a host where to send a frame to reach a particular IP address





Default Router

IP = 1.2.3.4

EUI-48 = AA-BB-CC-DD-EE

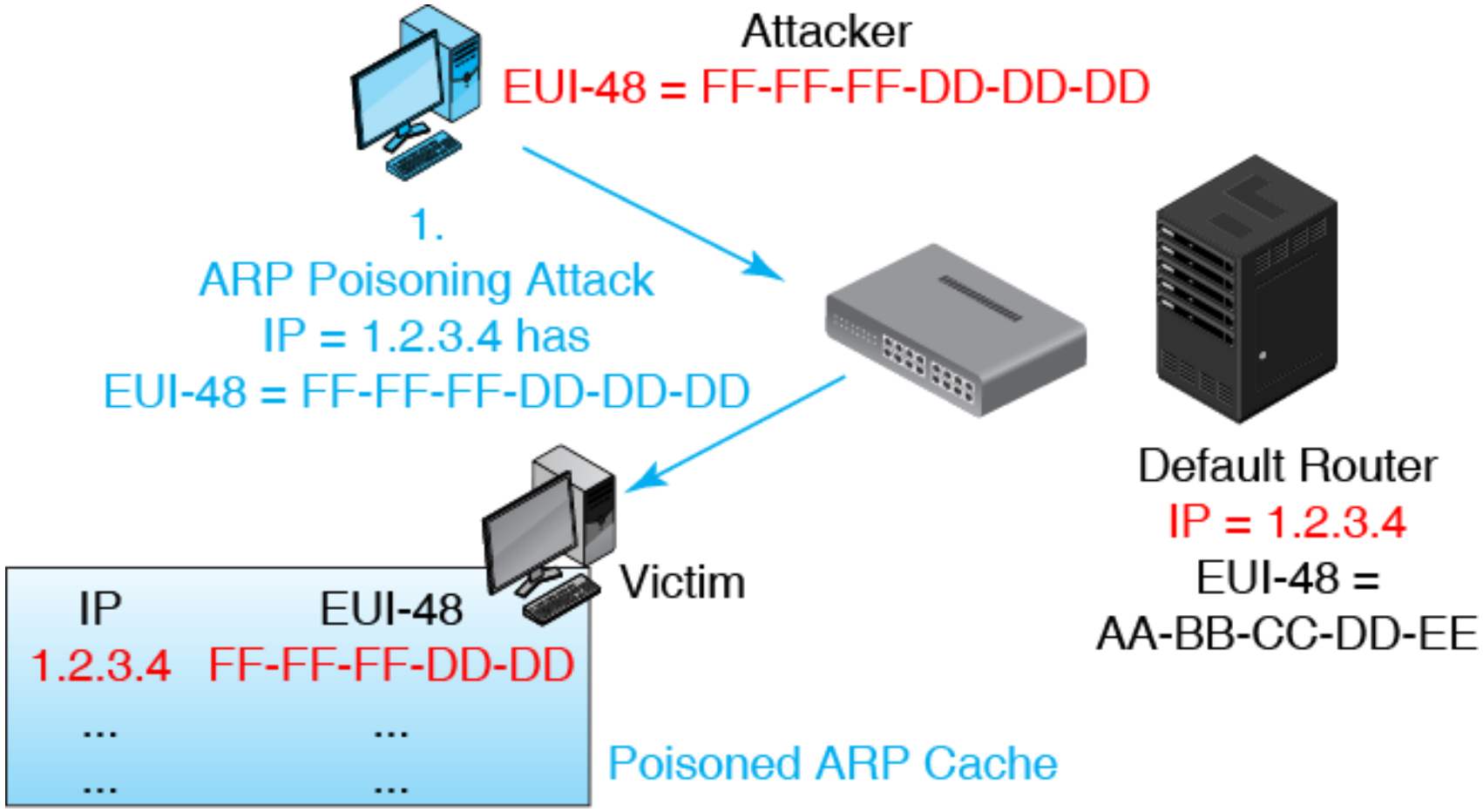


Victim

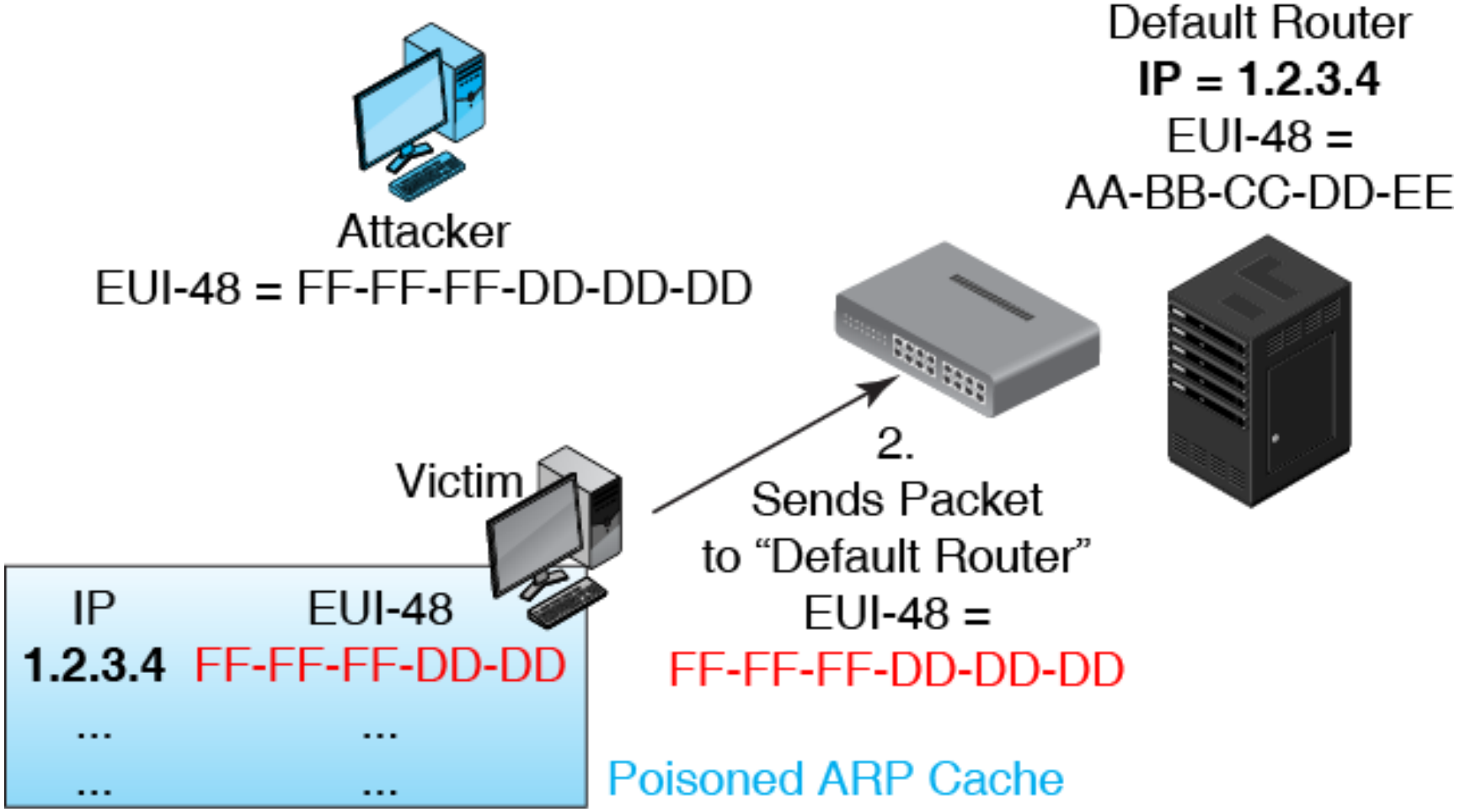
IP	EUI-48
1.2.3.4	AA-BB-CC-DD-EE
...	...
...	...

Correct ARP Cache

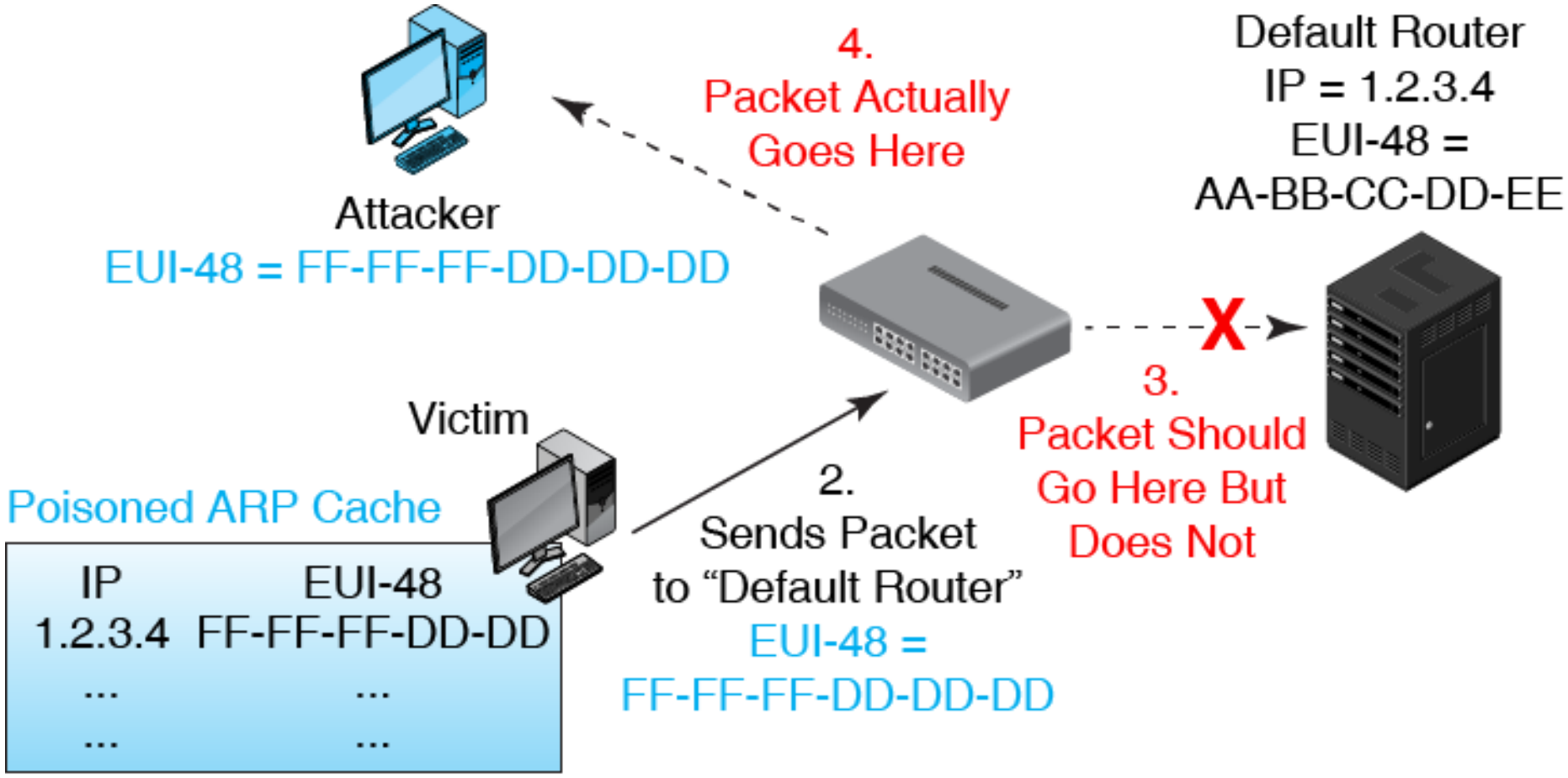
5.26 Man in the Middle Attack in an Ethernet LAN Using ARP Poisoning (2/4)



5.26 Man in the Middle Attack in an Ethernet LAN Using ARP Poisoning (3/4)



5.26 Man in the Middle Attack in an Ethernet LAN Using ARP Poisoning (4/4)



Outline of Chapter 5

- Ethernet Begins
- Introduction
- Physical Layer Standards
- Data Link Layer Standards
- Advanced Switch Operation
- **Ethernet Security**